Role of business formation and exit for unemployment and vacancy dynamics

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Abstract

How do business formation and exit dynamics influence job creation and destruction? We develop a unified model of endogenous business formation and exit and labor market frictions and study its implications for unemployment and vacancy dynamics. Monopolistically competitive producers use labor and face exit risk from product obsolescence and stochastic fixed costs. Creating businesses and vacancies both require sunk costs, making them partially predetermined: if a product line survives, unfilled vacancies are reposted. We estimate the model via Bayesian simulated method of moments using aggregate technology, product destruction, and match separation shocks. When a firm exits, it immediately eliminates jobs and posted vacancies and reduces product variety, so vacancy persistence closely tracks product-line longevity. By contrast, match separation shocks destroy only filled jobs, generating counterfactual positive comovement of unemployment and vacancies. The loss of variety also dampens job creation and employment via an aggregate demand externality. Adverse technology shocks induce exit among high-cost firms, amplifying labor-market effects. Quantitatively, the model matches the volatility, autocorrelation, and cross-correlation of unemployment, vacancies, business formation, employment-to-unemployment transitions, and establishment exit, with technology and destruction shocks playing the predominant roles.

1. Introduction

Labor market variables do not recover at equal speed, and the disparity does not just depend on the severity of the initial shock. The 2020 recession produced the highest peak unemploy-

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ment rate of any postwar episode — 13.0 percent — yet labor markets had recovered within two years, accumulating only 26.6 percentage-point quarters of total unemployment excess over twenty quarters from recession onset. The 2008–09 recession generated a smaller peak unemployment rate of 9.9 percent yet produced 76.6 percentage-point quarters of cumulative damage and a recovery stretching across six years. The standard Diamond–Mortensen–Pissarides (DMP) framework, in which all separations are identical and firms repost vacancies freely after any job loss, offers no structural account of why two recessions of comparable peak unemployment can generate such different cumulative damage: the composition of job loss — not just its volume — has no influence on persistence. We argue that a key missing margin is firm destruction — the permanent elimination of business opportunities and their associated vacancy slots — which ran at 7.8 percent of base employment in 2008–09 but only 2.9 percent in 2020.[2] When a firm exits, its vacancies vanish permanently and must be rebuilt from scratch at sunk cost; recovery requires not merely a rebound in profitability but a costly reconstruction of the vacancy stock, generating a persistent outward shift of the Beveridge curve. Provided vacancy continuation is sufficiently cheap relative to initial creation, surviving firms repost after separation, so the aggregate vacancy stock partially self-corrects even while unemployment remains elevated, producing a transitory displacement along the Beveridge curve rather than a persistent outward shift.

Coles and Moghaddasi Kelishomi (2018) show that extending the DMP model with finitely elastic vacancy creation enables separation shocks to generate a downward-sloping Beveridge curve and, jointly with technology shocks, fit a broad set of labor market moments well. But CK assume a single separation shock, conflating firm destruction and match dissolution into one margin. Gabrovski and Silva (2025) build on CK by distinguishing between destruction of a product line δ and breakdown of a match s, and show that under technology shocks alone, for empirically relevant values of δ, the model generates mild unemployment–labor productivity comovement and matches other key labor market moments, including the Beveridge curve. In both frameworks, sunk entry costs make the vacancy stock predetermined: v_{t-1} enters the equilibrium alongside u_t, and surviving firms repost vacancies freely following match dissolution,

[2]Financial frictions, nominal rigidities, and other margins absent from the benchmark model have each received substantial attention in the literature; we focus on firm destruction and its interaction with endogenous business formation.

substantially weakening the link between unemployment and contemporaneous productivity.

This paper opens additional channels related to business formation and exit. First, we introduce love-of-variety preferences over differentiated product lines, so that the number of active product lines N_t becomes a third independent state variable: greater product diversity raises the return to posting vacancies, while greater employment raises the resources available for product-line formation. We segment the product and labor markets so that producers can employ a mass of workers, avoiding any mechanical link between employment and the range of product lines: u_t, v_{t-1}, and N_t are jointly determined in equilibrium without an accounting restriction. Second, we endogenize firm exit, so that δ_t responds to aggregate productivity and separation conditions: a recession that reduces profitability or raises separation rates induces higher exit, further depleting the vacancy stock and product variety and further suppressing job creation. Together these extensions sharpen the asymmetry between δ and s shocks: a firm-destruction shock simultaneously damages all three state variables — raising u_t, depleting v_{t-1}, and eliminating product lines that would otherwise sustain vacancy creation incentives — while a match-separation shock raises u_t but does not directly affect v_{t-1} or N_t. We argue quantitatively that this three-channel amplification helps account for the cross-recession gap in cumulative labor market damage, and we identify the key structural parameters using Bayesian simulated method of moments on unconditional moments, extended with Bayesian impulse response matching to an externally identified firm-destruction shock.

Three empirical facts motivate treating firm destruction as a distinct and quantitatively important margin. *First, establishment closings and openings constitute a large and cyclically sensitive component of gross job flows.* Table 1 extends Table 2 in Jaimovich and Floetotto (2008) through 2024Q4 using Business Economics Database (BED) data. Columns 1–2 of Table 1 report the average fraction of quarterly gross job gains (losses) accounted for by opening (closing) establishments. Columns 3 and 4 report the ratio of the cyclical standard deviation of entry/exit flows to that of total flows, using the Hamilton (2018) regression filter ($h = 8, p = 4$) applied to raw levels. Columns 5 and 6 repeat the calculation with the HP filter ($\lambda = 1600$), which closely reproduces Jaimovich and Floetotto (2008)'s benchmark of approximately 0.33 in the original 1992–2006 sample.

Columns 1 and 2 show that closing establishments account for 19.3 percent of gross job losses on average across all private-sector supersectors — a share that is remarkably stable across industries ranging from construction (20.5 percent) to financial activities (24.6 percent). The relative standard deviation columns show that closing-establishment flows are meaningfully cyclical under both Hamilton and HP filters, with the total-private closing share moving at 0.152 times the cyclical variation of unemployment under the Hamilton filter. Translating these shares into rates, the quarterly establishment exit rate averages 0.93 percent of base employment against a total quarterly separation rate of 9.3 percent: most separations are therefore not firm exits, which makes the δ-vs-s decomposition empirically tractable — the two margins are separately measurable yet distinct enough that conflating them materially distorts model predictions.

	Mean share		Rel. std dev (Ham)		Rel. std dev (HP)	
Industry	$\bar{g}^O$	$\bar{g}^C$	σ^O/σ	σ^C/σ	σ^O/σ	σ^C/σ
Total private	0.199	0.193	0.227	0.152	0.206	0.132
Natural resources & mining	0.163	0.156	0.281	0.177	0.261	0.176
Construction	0.208	0.205	0.188	0.185	0.209	0.165
Manufacturing	0.123	0.137	0.214	0.109	0.219	0.093
Wholesale trade	0.203	0.226	0.251	0.184	0.260	0.143
Retail trade	0.169	0.152	0.202	0.142	0.210	0.124
Transportation & utilities	0.163	0.169	0.236	0.118	0.194	0.097
Information	0.206	0.209	0.284	0.310	0.301	0.269
Financial activities	0.235	0.246	0.343	0.323	0.416	0.305
Prof. & business services	0.205	0.211	0.242	0.156	0.236	0.133
Education & health services	0.203	0.211	0.307	0.151	0.246	0.125
Leisure & hospitality	0.241	0.196	0.173	0.168	0.174	0.158
Other services	0.225	0.223	0.427	0.286	0.415	0.270

Notes. $\bar{g}^O$ ($\bar{g}^C$): mean share of gross job gains (losses) from opening (closing) establishments, $\sum G_O/\sum G$. Cols 3–4: relative std dev σ^O/σ (σ^C/σ) via Hamilton (2018) filter (h=8, p=4), raw levels; robust to GFC/Covid outliers. Cols 5–6: same statistic via HP filter (λ=1600).

Table 1: Job Gains and Losses from Opening and Closing Establishments (*Extended to 2024Q4 (BED end)*) *Source: BLS Business Employment Dynamics (BED), quarterly, 12 Bartik supersectors. Replication and extension of Jaimovich & Floetotto (2008, JME) Table 2.*

Second, cumulative establishment exit predicts the severity and persistence of labor market disruption across recessions. Figure 1 plots the permanence-adjusted establishment exit rate against cumulative unemployment and vacancy gaps across the three post-1992 NBER recessions. The 2008–09 recession generated the largest exit rate (7.8% of base employment) and the largest cumulative unemployment gap (76.6 pp-quarters), reflecting both a deep trough and a protracted recovery. The 2001 recession, despite a more moderate exit rate (5.6%), produced the largest cumulative vacancy shortfall (25.7 pp-quarters) of the three episodes — exceeding even the GFC (18.8 pp-quarters) — driven by an unusually prolonged vacancy formation freeze following the collapse of investment demand after the dot-com bust, a pattern that likely reflects demand-side forces beyond establishment exit alone. The 2020 recession, with the smallest exit rate (2.9%), generated rapid recovery in both labor market margins and negligible cumulative vacancy shortfall (1.7 pp-quarters), consistent with a supply disruption that left the matching technology largely intact. Across these three episodes, larger establishment exit is associated with greater and more persistent labor market damage, particularly in the unemployment dimension.

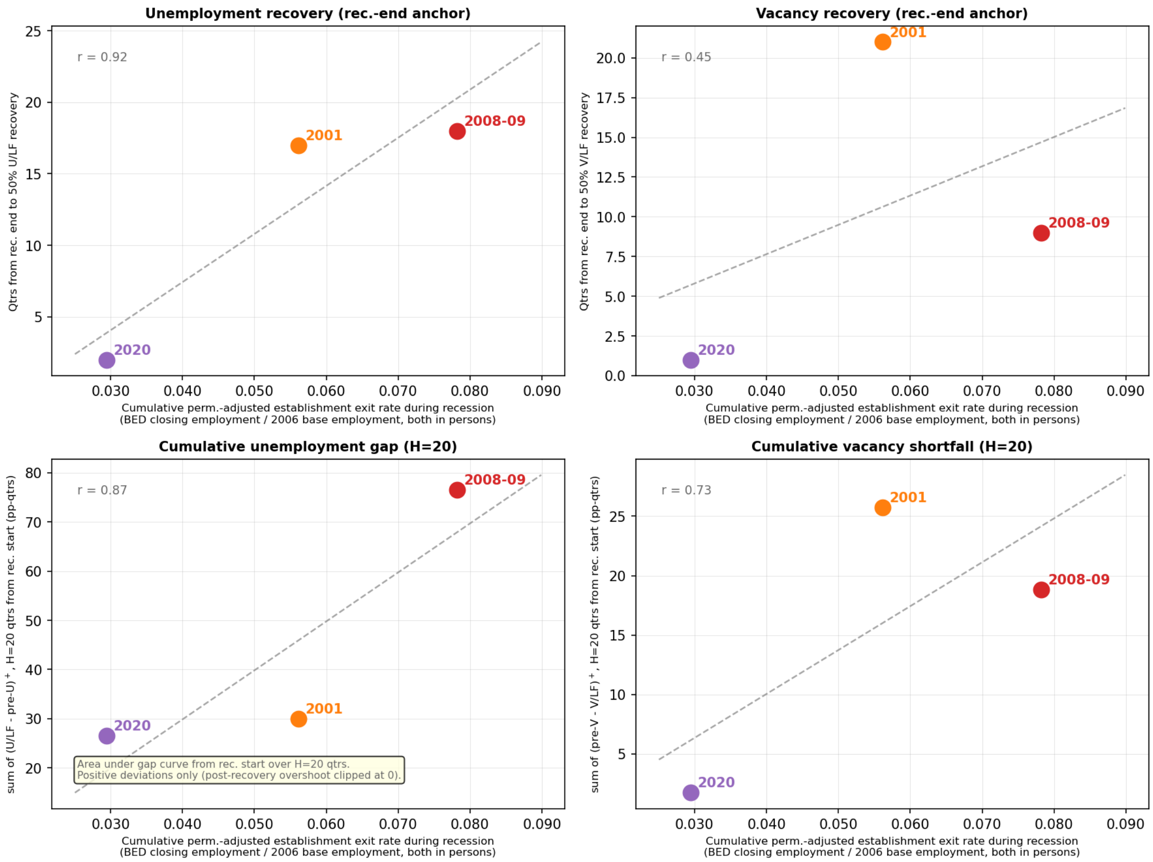

Figure 1: Relationship between cumulative establishment exit rates (BED Deaths) during recessions and labor market outcome measures. The top row shows quarters from the NBER recession end date until unemployment (left) and vacancy (right) rates recover 50% of the trough gap. The bottom row shows the cumulative unemployment gap (CUG, left) and cumulative vacancy shortfall (CVS, right), defined as the sum of positive deviations from the pre-recession baseline over $H = 20$ quarters from recession onset. Exit rates are constructed from BLS Business Employment Dynamics (BED) establishment deaths (establishments absent for four or more consecutive quarters), normalized by 2006 base-year employment.

Third, states with structurally higher establishment exit rates accumulate larger labor market gaps during recessions. Figure 2 plots each state's full-sample average Bartik-weighted establishment exit rate against its cumulative unemployment gap (CUG) and cumulative vacancy shortfall (CVS), both summed over 20 quarters from recession onset and averaged across the 2001, 2007–09, and 2020 recessions. The labor-force-weighted cross-sectional correlations are 0.65 (CUG) and 0.28 (CVS): states with structurally higher exit propensity accumulate substantially more total unemployment damage during downturns, with the vacancy pattern weaker but directionally consistent. The stronger CUG correlation is itself informative — it suggests that exit intensity shapes the *duration* of labor market distress more than the *depth* of the vacancy shortfall, consistent with the model's prediction that firm destruction generates persistent Beveridge curve dislocations rather than transitory displacements.[3] However, the cross-sectional pattern conflates structural differences in industrial composition with the causal effect of exit shocks: high-exit states may differ from low-exit states along dimensions — industry mix, institutional environment, average firm age — that independently predict labor market performance. To isolate causal transmission, we construct Bartik shift-share instruments that exploit within-state variation in exit exposure driven by national industry-level closing rates rather than local conditions, building separate instruments for δ from BED establishment closings and for s from JOLTS layoff and discharge rates. We then estimate local projection impulse response functions to externally identify the dynamic transmission of each shock to state-level unemployment and vacancies, providing a quasi-experimental foundation for the structural parameters prior to Bayesian estimation.

[3]The weaker CVS correlation is also consistent with the model: the vacancy stock partially self-corrects through reposting by surviving firms, attenuating the cross-state vacancy signal relative to the unemployment signal.

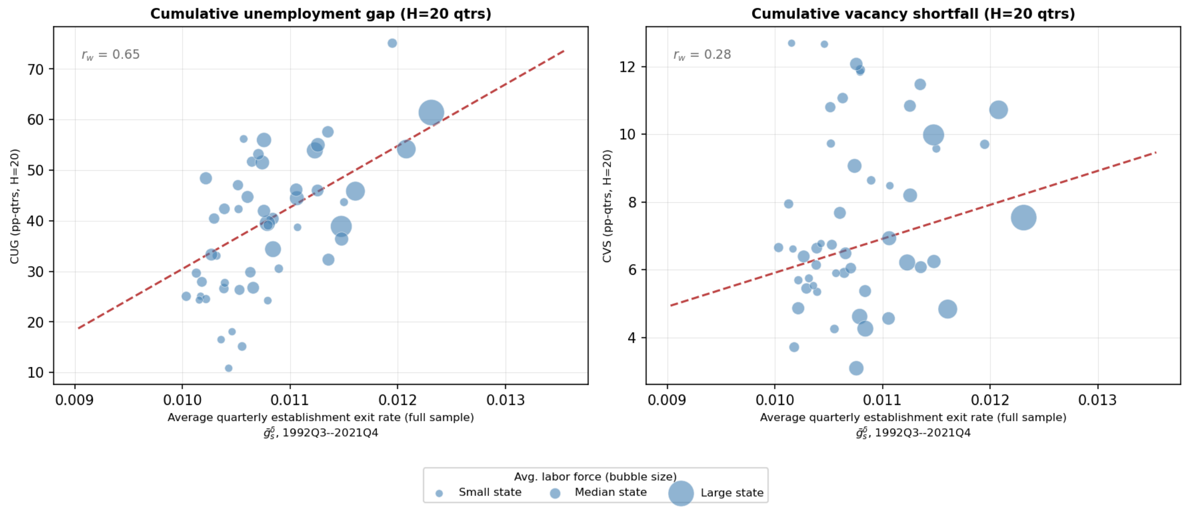

Figure 2: Cross-state relationship between average establishment exit rates and cumulative recession labor market damage. Each point is one of the 50 U.S. states; bubble size is proportional to average labor force. The horizontal axis is the full-sample time average of the state-level Bartik-weighted establishment exit rate ($\bar{g}^{\delta}_{s}$, 1992Q3–2021Q4), constructed from BLS Business Employment Dynamics (BED) data using 2006 base-year employment shares. The left panel shows the cumulative unemployment gap (CUG): the sum of quarterly deviations of the unemployment rate above its pre-recession average over $H = 20$ quarters from recession onset, in percentage-point-quarters. The right panel shows the analogous cumulative vacancy shortfall (CVS) for the vacancy rate. Both outcomes are averaged across the three post-1992 NBER recessions (2001, 2007–2009, 2020), weighted by state labor force at recession onset. The regression line is estimated by WLS with state labor force weights; r_w denotes the labor-force-weighted correlation.

Following Goldsmith-Pinkham, Sorkin, and Swift (2020), we bring quasi-experimental discipline to the δ transmission channel using Bartik shift-share instruments constructed from BED establishment deaths — the subset of closings absent from the UI payroll base for four or more consecutive quarters, aligning the instrument directly with the model's concept of irreversible product-line destruction. The instrument weights leave-one-out national industry-level death rates by each state's base-year (2006) industry employment shares. The identifying variation is geographic: a state with high pre-recession exposure to industries experiencing national waves of establishment closings — manufacturing in 2008–09, for instance — faces a large exit shock driven by national industry conditions rather than local demand, and can be compared to low-exposure states in the same quarter to isolate the causal labor market effect of firm destruction. The instrument is residualized on aggregate productivity growth to address cross-instrument correlation arising mechanically during aggregate downturns.

We estimate panel local projections (Jordà, 2005) with state and time fixed effects, standard errors clustered at the state level, and outcomes capped at 2019Q4 to exclude the Covid episode. The δ instrument generates a persistent decline in both unemployment and vacancies: the unemployment response peaks at +1.57 percentage points at $h = 19$ quarters and remains significant throughout $h = 0$–20; the vacancy response falls monotonically, reaching −0.426 log points at $h = 14$ and remaining significant from $h = 1$ onward. Both patterns are consistent with the model's prediction that firm destruction generates a persistent outward shift of the Beveridge curve rather than a transitory displacement along it. The estimated $\delta \to v$ and $\delta \to u$ coefficients approximate the aggregate transmission elasticities and provide external moment conditions for Bayesian impulse response matching, combined with unconditional moments in the Bayesian simulated method of moments estimation described in Section 5.[4]

The model is estimated by Bayesian simulated method of moments, combining two sets of moment conditions. The first is closely related to Coles and Moghaddasi Kelishomi (2018): unconditional moments of unemployment, vacancies, labor productivity, and aggregate job finding and separation rates, augmented with the mean and relative volatility of the permanent establishment exit rate (BED Deaths), which disciplines the level and cyclical variation of the destruction process directly rather than inferring it residually from labor market outcomes. The second set consists of the empirically identified $\delta \to u$ and $\delta \to v$ impulse responses from the Bartik local projections, matched via Bayesian impulse response matching with inverse-variance weights. The two sets are complementary: unconditional moments pin down the destruction process and the steady-state labor market, while the conditional moments discipline the structural transmission parameters governing how costly and slow vacancy replacement is following a firm destruction shock.[5] Quantitative results and the two-model counterfactual — comparing the full model's implied recovery paths to those of a Coles-Kelishomi limiting case that abstracts from variety effects and endogenous exit — are reported in Section 5.

[4]The δ unemployment response concentrates almost entirely in states experiencing tight financial conditions: at average financial conditions the point estimate is near zero and insignificant at all horizons beyond $h = 0$. This state-dependence in the unemployment channel, absent from the baseline model, is consistent with financial amplification of firm destruction and is left for future work; the vacancy channel, which enters the structural estimation as a moment condition, does not exhibit the same state-dependence.

[5]The unconditional moments identify the level and volatility of the δ process, while the conditional moments identify the transmission parameters ξ and x_m governing the vacancy creation cost distribution. A model with the same δ process but different ξ would match the former and fail the latter.

This paper contributes to four strands of literature. The first is search-and-matching models with finitely elastic vacancy creation, where our model builds most directly on Coles and Moghaddasi Kelishomi (2018) and Gabrovski and Silva (2025); Broer, Druedahl, Harmenberg, and Öberg (2025) independently combine finitely elastic vacancy creation with endogenous separation but abstract from product variety and the separate destruction margin. The second is endogenous product variety and firm dynamics in business cycles, where we build on Bilbiie, Ghironi, and Melitz (2012) and depart from the one-firm-one-worker structure of Shao and Silos (2013); Colciago, Fasani, and Rossi (2021) combine endogenous entry with labor market frictions in a related framework but maintain infinitely elastic vacancy creation and abstract from the destruction margin. The third is Bartik shift-share identification in local labor markets (Goldsmith-Pinkham, Sorkin, and Swift, 2020; Nakamura and Steinsson, 2014), discussed alongside the instrument construction in Section 4. The fourth is Bayesian structural estimation of search models, extending the simulated method of moments approach of Coles and Moghaddasi Kelishomi (2018) to incorporate impulse response matching as an additional set of moment conditions (Christiano, Eichenbaum, and Trabandt, 2016). The remainder of the paper is organized as follows. Section 4 presents the motivating evidence and Bartik local projections. Section 2 develops the model environment. Section 3 derives the equilibrium, establishes the δ-vs-s asymmetry formally, and characterizes steady-state properties. Section 5 presents calibration, estimation, and the two-model counterfactual. Section **??** concludes.

2. Environment

Time is discrete and infinite. The economy is populated by households, intermediaries (recruiters), and retailers. Households and recruiters each have unit mass; the mass of retailers is endogenous, each supplying one product line. A household consists of a large family of workers who are either employed or unemployed, as in Merz (1995). Employed members supply one unit of labor inelastically. The household consumes and purchases equity. A unit measure of recruiters can each create one vacancy at a time, at a sunk cost drawn from a known distribution, with each vacancy specialized to a single product line. Retailers rent labor services from matched recruiters and produce the specialized variety.

Each period a number $m(u_t, v_t)$ of recruiter–worker pairs form, given unemployment u_t and vacancies v_t. We adopt the Cobb-Douglas matching function $m(u, v) = Au^{\eta_L}v^{1-\eta_L}$, where η_L

is the elasticity of matches with respect to unemployment (Petrongolo and Pissarides, 2001). Constant returns to scale implies that the job-finding rate $f(\theta) = A\theta^{1-\eta_L}$ and the vacancy-filling rate $q(\theta) = A\theta^{-\eta_L}$ depend only on market tightness $\theta \equiv v/u$.

2.1. Preferences

We develop the model with general homothetic preferences as in Bilbiie, Ghironi, and Melitz (2012), then apply specific functional forms in the quantitative analysis. The household consumes a basket c_t, a homothetic aggregate over a continuum of goods Ω. The welfare-based price index P_t takes only individual prices as arguments, and demand for variety j satisfies

$$c_{jt} = c_t \partial P_t / \partial p_{jt}. \tag{1}$$

The symmetric price elasticity of demand is in general a function of the measure N_t of goods:

$$\varepsilon(N_t) = -\frac{\partial \log c_j}{\partial \log p_j}.$$

The welfare gain from an additional product variety is captured by the relative price and its elasticity:

$$\rho_j = \rho(N) \equiv \frac{p_j}{P}, \quad \zeta(N) = \frac{\rho'(N)}{\rho(N)} N.$$

The relative price rises with variety: with more products available at the same nominal price, the welfare-based cost of living falls, so ρ increases. Equivalently, spreading consumption across more goods benefits the household. The two statistics $\varepsilon(N_t)$ and $\rho(N_t)$ completely characterize symmetric homothetic preferences in the model.

Under standard Dixit–Stiglitz preferences (DS-CES), ε measures both the taste for diversity and the elasticity of substitution, so both the markup and the variety benefit are independent of the number of producers: $\zeta(N) = \mu(N) - 1 = 1/(\varepsilon - 1)$. The second specification is generalized CES following Benassy (1996), with consumption basket $N^{\zeta - 1/(\varepsilon-1)} \left(\int c_j^{(\varepsilon-1)/\varepsilon} dj \right)^{\varepsilon/(\varepsilon-1)}$; this disentangles the taste for variety ζ from the elasticity of substitution ε. The third specification uses the translog expenditure function of Feenstra (2003), under which $\varepsilon(N) = 1 + \eta N$, so goods become more substitutable as variety increases. Table 2 summarizes key statistics for all three specifications.

	DS-CES	General CES	Translog
Consumption index	$\left(\int c_j^{(\varepsilon-1)/\varepsilon} dj\right)^{\varepsilon/(\varepsilon-1)}$	$N^{\zeta-1/(\varepsilon-1)} \left(\int c_j^{(\varepsilon-1)/\varepsilon} dj\right)^{\varepsilon/(\varepsilon-1)}$	–
Price index	$\left(\int p_j^{1-\varepsilon} dj\right)^{1/(1-\varepsilon)}$	$N^{-\zeta+1/(\varepsilon-1)} \left(\int p_j^{1-\varepsilon} dj\right)^{1/(1-\varepsilon)}$	–
Markup $\mu(N)$	$\frac{\varepsilon}{\varepsilon-1}$	$\frac{\varepsilon}{\varepsilon-1}$	$1 + \frac{1}{\eta N}$
Relative price $\rho(N)$	$N^{\frac{1}{\varepsilon-1}}$	N^{ζ}	$\exp\left\{-\frac{1}{2}\frac{\tilde{N}-N}{\eta\tilde{N}N}\right\}$
Benefit of variety $\zeta(N)$	$\mu - 1$	ζ	$\frac{1}{2\eta N} = \frac{\mu(N)-1}{2}$

Table 2: Preference specifications: indices, markup, relative price, and benefit of variety

2.2. Laws of motion for firms, unemployment, and vacancies

At the start of period t, exogenous destruction δ_{t-1}, realized at the end of $t-1$, and the endogenous continuation-cost draw together determine which product lines are active. Specifically, a product line survives into period t if it was not destroyed exogenously at the end of $t-1$ (probability $1-\delta_{t-1}$) *and* its owner draws a continuation cost below the equilibrium cutoff x_t^c at the start of t (probability $F(x_t^c)$). Both events occur before production and matching, so u_t is unemployment *after* firm exit but *before* matching. The law of motion for active product lines is

$$N_t = (1-\delta_{t-1})F(x_t^c)(N_{t-1} + N_{t-1}^e) \tag{2}$$

where N_{t-1}^e denotes last-period entrants. The entry lag means a fraction of entrants are destroyed before they produce. The total exit rate combining exogenous destruction and endogenous exit is

$$\delta_{e,t} \equiv 1 - (1-\delta_{t-1})F(x_t^c), \tag{3}$$

where δ_t denotes the exogenous AR(1) destruction shock and $\delta_{e,t}$ denotes the total exit rate; we maintain this distinction throughout. The aggregate separation rate flowing into pre-matching unemployment u_t is

$$\tau_t = \delta_{e,t} + s_{t-1}(1-\delta_{e,t}), \tag{4}$$

where s_{t-1} is the match-separation shock that realized at end of $t-1$. The unemployment law of motion is

$$u_t = (1-f(\theta_{t-1}))u_{t-1} + \tau_t\big[(1-u_{t-1}) + f(\theta_{t-1})u_{t-1}\big] \tag{5}$$

The first term is last period's unemployed who did not find a job; the second is the fraction τ_t of last period's employed (including those who matched in $t-1$) who separate before production in t. This has the same form as the textbook model, with the total separation rate τ_t replacing the constant separation rate of DMP.

Sunk vacancy creation costs render vacancies partially predetermined: current vacancies evolve as

$$v_t = (1-\delta_{e,t})[(1-q(\theta_{t-1}))v_{t-1} + s_{t-1}(1-u_{t-1})] + e_t \tag{6}$$

Current vacancies are the sum of three components: (i) unmatched vacancies from last period that survived product destruction; (ii) filled jobs that experienced a match separation at end of $t-1$ but whose firm survived both margins, reposted at no additional sunk cost; and (iii) vacancies created by new entrants e_t, who enter at the start of t before matching. Note that $(1-\delta_{e,t})$ multiplies both (i) and (ii): only firms surviving both exogenous destruction and the continuation-cost draw contribute to either term, so a shock to δ_t collapses the entire bracket simultaneously. The asymmetric implications of these laws of motion for the transmission of δ and s shocks are derived formally in Section 3.

2.3. Stochastic processes

Technology, product destruction, and idiosyncratic separation each follow an AR(1) process in logs:

$$\log z_t = \rho_z \log z_{t-1} + (1-\rho_z)\log \bar{z} + e_{zt} \tag{7}$$

$$\log \delta_t = \rho_\delta \log \delta_{t-1} + (1-\rho_\delta)\log \bar{\delta} + e_{\delta t}$$

$$\log s_t = \rho_s \log s_{t-1} + (1-\rho_s)\log \bar{s} + e_{st} \tag{8}$$

where $\bar{z}$, $\bar{\delta}$, and $\bar{s}$ are the long-run means of technology, exogenous product destruction, and idiosyncratic separation, respectively, and innovations e_{zt}, $e_{\delta t}$, e_{st} are mutually uncorrelated. Shocks s_t and δ_t realise at the end of period t, after production, bargaining, and matching, as illustrated in Figure 3. Agents making vacancy and entry decisions at the start of t therefore act on s_{t-1} and δ_{t-1} — the shocks that realised at end of $t-1$ — which determines the total exit rate $\delta_{e,t} = 1-(1-\delta_{t-1})F(x_t^c)$ and aggregate separation rate $\tau_t = \delta_{e,t} + s_{t-1}(1-\delta_{e,t})$. This lag, combined with sunk vacancy creation costs, is the source of predetermination: vacancies and employment are partially predetermined with respect to current shocks. Here is the complete timing figure code: latex

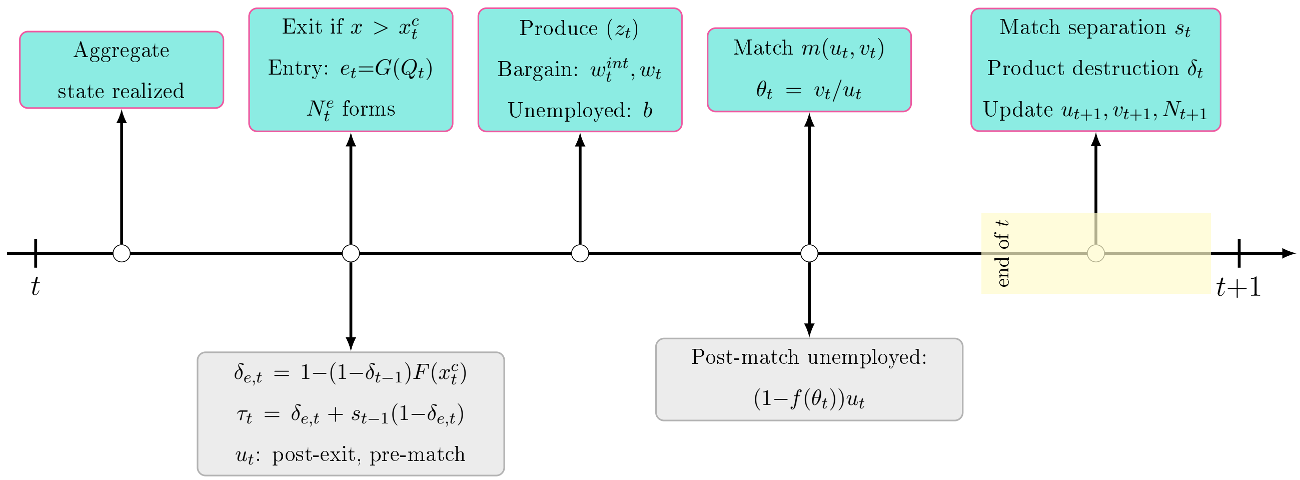

Figure 3: Timing. Turquoise: agent decisions. Grey: implied states. Yellow band: shocks realizing end of t, governing $t+1$.

3. Equilibrium

This section derives equilibrium given the environment of Section 2. Table 3 positions the paper relative to the closest antecedents across four model dimensions. Equilibrium comprises three blocks: a business formation block governing the evolution of product lines and firm values, a labor market block governing vacancy creation, matching, and wage determination, and stochastic processes for the three structural shocks. Section ?? derives formally the asymmetric transmission of δ and s shocks that motivates the paper's empirical strategy.

	Finitely elastic vacancies	Endogenous variety	Endogenous separation	Separate destruction margin
Bilbiie, Ghironi, and Melitz (2012)	NA	✓	NA	NA
Coles and Kelishomi (2018)	✓	×	×	×
Broer, Druedahl, Harmenberg, and Öberg (2025)	✓	×	✓	×
Gabrovski and Silva (2025)	✓	×	×	✓
Shao and Silos (2013)	✓	✓	×	×
Christiano, Eichenbaum, and Trabandt (2016)	×	×	×	×
Colciago, Fasani, and Rossi (2021)	×	✓	×	×
Cacciatore and Fiori (2016)	×	✓	✓	×
This paper	✓	✓	✓	✓

Table 3: Comparison of model ingredients. *Finitely elastic vacancies*: sunk vacancy creation costs generating a positive vacancy value. *Endogenous variety*: equilibrium number of product lines responds to economic conditions. *Endogenous separation*: firm exit driven by idiosyncratic profitability draws. *Separate destruction margin*: exogenous product-line destruction distinguished from match separation.

3.1. Household problem

Let ν_t denote the ex-dividend equity price of the aggregate firm portfolio and ν_t^f the post-dividend value of an individual firm, following Bilbiie, Ghironi, and Melitz (2012). Given risk-aversion parameter σ and discount factor β, the household chooses consumption c_t and equity shares a_t to maximize expected discounted utility:

$$\max_{c_t,\, a_t} \sum_{t=0}^{\infty} \mathbb{E}_0\, \beta^t c_t^{1-\sigma}/(1-\sigma)$$

subject to

$$c_t + \nu_t a_{t+1} = w_t(1-u_t) + bu_t + (\nu_t + d_t)a_t + D_t^{int} - T_t \tag{9}$$

where b is the value of unemployment, d_t is the per-share dividend, D_t^{int} are intermediary dividends, and T_t are lump-sum taxes financing unemployment benefits. Since households have unit mass, $c_t = C_t$ in equilibrium. The first-order condition for equity shares yields the standard asset-pricing equation:

$$\lambda_t \nu_t = \beta\, \mathbb{E}_t \lambda_{t+1}[\nu_{t+1} + d_{t+1}] \tag{10}$$

where $\lambda_t = c_t^{-\sigma}$ is the marginal utility of consumption and $m_{t+1} \equiv \beta(\lambda_{t+1}/\lambda_t)$ is the stochastic discount factor. To express (10) at the firm level, use the mappings $\nu_t = \nu_t^f(N_t + N_t^e)$ and

$\nu_t + d_t = (\nu_t^f + d_t^f)N_t$, together with the firm law of motion (2). This yields the firm-level Euler equation, familiar from BGM and extended here to incorporate endogenous exit:

$$\nu_t^f = \mathbb{E}_t m_{t+1}(1-\delta_t)F(x_{t+1}^c)(\nu_{t+1}^f + d_{t+1}^f) \tag{11}$$

Equation (11) equates the current firm value to the expected discounted sum of next-period firm value and dividends, adjusted for the stochastic discount factor m_{t+1} and the survival probability $(1-\delta_t)F(x_{t+1}^c)$, which combines the probability of surviving exogenous destruction $(1-\delta_t)$ and the probability of drawing a continuation cost below the equilibrium cutoff $F(x_{t+1}^c)$.

3.2. Recruiters

There is a fixed unit measure of recruiters who can create vacancies subject to a sunk cost x drawn from cdf $G(\cdot)$. Additionally, if a vacancy is filled the recruiter pays a fixed real cost κ before bargaining with the newly matched worker.[6] After matching, the recruiter rents labor to an incumbent retailer with product line j. We introduce recruiters as a separate agent type to avoid the complications of Stole-Zwiebel intra-firm bargaining that arise when workers negotiate directly with retailers facing a downward- sloping demand curve: with a separate recruiter, the bilateral bargaining problem reduces to the standard DMP surplus-splitting equation regardless of the retail market structure.[7] Let Q_t^j and J_t^j denote the values of an unfilled and filled vacancy associated with product line j.

The value of an unfilled vacancy for product line j satisfies

$$\begin{aligned} Q_t^j &= \mathbb{E}_t m_{t+1}(1-\delta_t)F(x_{t+1}^c)\left[(1-q(\theta_t))Q_{t+1}^j + q(\theta_t)J_{t+1}^j\right] - q(\theta_t)\kappa \\ &= \mathbb{E}_t m_{t+1}(1-\delta_t)F(x_{t+1}^c)\left[Q_{t+1}^j + q(\theta_t)(J_{t+1}^j - Q_{t+1}^j)\right] - q(\theta_t)\kappa \end{aligned} \tag{12}$$

An unmatched recruiter survives to next period only if the product line survives both exogenous destruction and the continuation-cost draw, with probability $(1-\delta_t)F(x_{t+1}^c)$. In the next period, the vacancy fills with probability $q(\theta_t)$, generating value J_{t+1}^j, or remains unfilled with probability

[6] Pissarides (2009) first suggested this matching cost to make vacancy posting more responsive; Christiano, Eichenbaum, and Trabandt (2016) estimate it accounts for the dominant share of recruiting costs.

[7] Under Stole-Zwiebel bargaining, retailers would strategically over-hire to improve their bargaining position, generating a fixed-point problem in employment that significantly complicates aggregation. The recruiter segmentation is isomorphic to the retailer– wholesaler structure of Christiano, Eichenbaum, and Trabandt (2016), with recruiters replacing intermediate goods.

$1-q(\theta_t)$. The matching cost κ is paid upon filling. The recruiter chooses the best product line: $Q_t \equiv \max_j Q_t^j$. In equilibrium, competition equalizes values across active product lines: $Q_t^j = Q_t$ for all j with $v_t^j > 0$.

Define the flow value of a vacancy as the net benefit of holding a vacancy this period relative to next:

$$K_t \equiv Q_t - \mathbb{E}_t m_{t+1}(1-\delta_t)F(x_{t+1}^c)Q_{t+1}$$

Rearranging (12) places the average hiring cost on the left-hand side:

$$\kappa + \frac{K_t}{q(\theta_t)} = \mathbb{E}_t m_{t+1}(1-\delta_t)F(x_{t+1}^c)(J_{t+1}^j - Q_{t+1}) \tag{13}$$

A recruiter opens a vacancy whenever $x \leq Q_t$, so total newly opened vacancies are $e_t = G(Q_t)$. Following CK, we adopt a power law distribution:

$$G(x) = \left(\frac{x}{x_m}\right)^{\xi}, \quad x \in [0,\, x_m]$$

with density $g(x) = \xi x^{\xi-1}/x_m^{\xi}$.[8] Provided $Q < x_m$, inverting $e = G(Q)$ gives

$$Q_t = e_t^{1/\xi}\, x_m$$

so the elasticity of the vacancy value with respect to entry is $1/\xi$. Free entry of vacancies ($\xi \to \infty$) nests the DMP baseline.

By symmetry of the retail problem (established in Section 3.4), recruiter compensation $w_t^{int,j}$ and the value of a filled vacancy J_t^j are identical across active product lines; we henceforth drop the superscript j. Recruiters supply labor in a competitive market, taking w_t^{int} as given. The value of a matched recruiter is[9]

$$J_t = w_t^{int} - w_t + \mathbb{E}_t m_{t+1}(1-\delta_t)F(x_{t+1}^c)\big(s_t Q_{t+1} + (1-s_t)J_{t+1}\big) \tag{14}$$

The recruiter receives w_t^{int} from the retailer and pays the worker w_t. Conditional on product-line survival, the match continues with probability $1-s_t$; otherwise the vacancy is reposted at no additional sunk cost.

[8] If $\xi > 1$ the density is rising (relatively more high-cost recruiters); if $\xi < 1$ it is falling; if $\xi = 1$ costs are uniform on $[0, x_m]$.

[9] The asset-pricing equations (12) and (14) can equivalently be obtained by having the household purchase shares of recruiter firms directly; the stochastic discount factor ensures equivalence. Writing it via the household problem makes the derivation more compact and facilitates comparison to BGM and Mortensen and Pissarides (1994).

Job creation condition.. Subtracting (12) from (14) and using the definition of K_t yields recruiter surplus:

$$J_t - Q_t = w_t^{int} - w_t - K_t + (1 - s_t)\left(\kappa + \frac{K_t}{q(\theta_t)}\right) \tag{15}$$

Substituting (15) into the right-hand side of (13) yields the job creation condition:

$$\kappa + \frac{K_t}{q(\theta_t)} = \mathbb{E}_t m_{t+1}(1-\delta_t)F(x_{t+1}^c)\left(w_{t+1}^{int} - w_{t+1} - K_{t+1} + (1 - s_{t+1})\left(\kappa + \frac{K_{t+1}}{q(\theta_{t+1})}\right)\right) \tag{16}$$

The left-hand side is the average hiring cost: the fixed matching cost κ plus the flow cost $K_t/q(\theta_t)$, equal to the opportunity cost of holding the vacancy this period multiplied by its expected duration $1/q(\theta_t)$. The right-hand side is the present discounted value of the match: future net revenue minus the future flow entry cost, plus the continuation value if the worker is retained, discounted by the stochastic discount factor and the product-line survival probability.

3.3. Wage determination

Nash bargaining between the recruiter and worker yields

$$w_t = \phi\left(w_t^{int} - K_t + f(\theta_t)\left(\kappa + \frac{K_t}{q(\theta_t)}\right)\right) + (1-\phi)b \tag{17}$$

where ϕ is the worker's bargaining power.[10] Recruiter compensation $w_t^{int} = \rho(N_t)z_t/\mu(N_t)$ — derived in Section 3.4 — enters negatively through the markup. The flow entry cost K_t has opposing effects: it reduces the match surplus (dampening wages by ϕK_t) but improves the worker's threat point, raising the tightness slope of the wage curve by $\phi\theta K_t$; the second effect dominates when $\theta > 1$. The fixed cost κ raises wages by $\phi f(\theta_t)\kappa$ because a failed negotiation forces the firm to pay κ again upon next filling the vacancy. Setting $w_t^{int} = z_t$ and $\kappa = 0$ recovers the CK wage equation.

Substituting (17) into (16) to eliminate w_t yields the wage-substituted job creation condition:

$$\begin{aligned}\kappa + \frac{K_t}{q(\theta_t)} = \mathbb{E}_t m_{t+1}(1-\delta_t)F(x_{t+1}^c)\Bigg[&(1-\phi)\left(w_{t+1}^{int} - K_{t+1} - b\right)\\ &- \phi\theta_{t+1}\left(K_{t+1} + q(\theta_{t+1})\kappa\right)\\ &+ (1 - s_{t+1})\left(\kappa + \frac{K_{t+1}}{q(\theta_{t+1})}\right)\Bigg]\end{aligned}$$

[10]The wage equation (17) has the same general form as Shao and Silos (2013), their equations 21 and 22; their setting includes capital affecting w_t^{int} and writes the wage using the vacancy value Q_t rather than the flow value K_t.

3.4. Retailers

A retailer selling variety j rents labor from recruiters at wage $w_t^{int,j}$. At the beginning of each period, before production, the firm draws a continuation cost $\chi \sim F_\chi$ in units of the consumption basket, where F_χ is a mixture between a mass point at zero with probability $1-p_0$ and a bounded power law:

$$F_\chi(\chi) = (1-p_0) + p_0\left(\frac{\chi}{\chi_m}\right)^\psi, \quad \chi \in [0,\, \chi_m]$$

A firm with draw $\chi > \chi_t^c$ exits; one with $\chi \leq \chi_t^c$ pays the cost and continues. Define the survival probability:

$$\Lambda_t \equiv F_\chi(\chi_t^c) = (1-p_0) + p_0\left(\frac{\chi_t^c}{\chi_m}\right)^\psi \tag{18}$$

so that the endogenous exit rate is $1-\Lambda_t = p_0\left(1-(\chi_t^c/\chi_m)^\psi\right)$, consistent with the total exit rate $\delta_{e,t} = 1-(1-\delta_{t-1})\Lambda_t$ from Section 2. The elasticity of the endogenous exit rate with respect to the cutoff is $\psi p_0 (\chi_t^c/\chi_m)^\psi/(1-\Lambda_t)$, which increases in ψ and falls as the mass point $1-p_0$ rises.

The value of a firm after its cost realization is:

$$\nu_t^f(\chi) = \max\left\{0,\ \max_{y_t}\left[\rho^d(y_t)y_t - w_t^{int}\frac{y_t}{z_t}\right] - \chi + (1-\delta_t)\mathbb{E}_t m_{t+1}\Lambda_{t+1}\,\nu_{t+1}^{ef}\right\} \tag{19}$$

where $\nu_t^{ef} \equiv (1/\Lambda_t)\int_0^{\chi_t^c} \nu_t^f(\tilde{\chi})\,dF_\chi(\tilde{\chi})$ is the expected firm value conditional on survival, and $\rho^d(y_t)$ is the inverse demand curve from (1), equal in symmetric equilibrium to $\rho(N_t)$. The profit maximization problem is static: letting $MR \equiv \rho'(y)y + \rho(y)$ denote marginal revenue, the first-order condition is

$$MR_t \cdot z_t = w_t^{int,j}$$

The left-hand side is the marginal revenue product; the right-hand side is recruiter compensation, which exceeds the worker's wage due to matching frictions. Equivalently, $MR_t = MC_t = w_t^{int}/z_t$. The standard markup condition $\rho_t = \mu(N_t)MC_t$ gives recruiter compensation:

$$w_t^{int} = \frac{\rho(N_t)}{\mu(N_t)}z_t$$

Recruiter compensation is increasing in the relative price $\rho(N_t)$ and technology z_t and decreasing in the markup $\mu(N_t)$. Since $\mu'(N_t) \leq 0$, w_t^{int} is increasing in N_t overall.[11] Retail profits gross of

[11] Following BGM, marginal cost is w_t^{int}/z_t. Under DS-CES, $\mu = \varepsilon/(\varepsilon-1)$ is constant, recovering the standard formula.

the continuation cost are a share $(\mu(N_t) - 1)/\mu(N_t)$ of sales:

$$R_t^f = \frac{\mu(N_t) - 1}{\mu(N_t)} \rho(N_t) y_t$$

3.5. Firm entry and exit

By definition $\nu_t^f(\chi_t^c) = 0$. Evaluating the firm Bellman (19) at $\chi = \chi_t^c$ yields the cutoff condition:

$$\chi_t^c = R_t^f + \nu_t^f \tag{20}$$

where $\nu_t^f \equiv \nu_t^{ef} \cdot \Lambda_t$ is the unconditional expected firm value. From the Bellman, the firm value conditional on the cost realization is $\nu_t^f(\chi) = \chi_t^c - \chi$ for all $\chi \leq \chi_t^c$, so that $\nu_t^{ef} = \chi_t^c - E(\chi \,|\, \chi \leq \chi_t^c)$. For the power law component, the conditional mean of a positive draw satisfies $E(\chi \,|\, 0 < \chi \leq \chi_t^c) = \psi_c \chi_t^c$ where $\psi_c \equiv \psi/(\psi + 1)$, so the overall conditional mean (including the mass point at zero) is $E(\chi \,|\, \chi \leq \chi_t^c) = p_0 \psi_c \chi_t^c$. Hence:

$$\nu_t^f = (1 - \delta_t)\mathbb{E}_t m_{t+1} \Lambda_{t+1} \, \nu_{t+1}^{ef}$$

New firms are created using technology $N_t^e = z_t l_t^e / f_e$, where l_t^e is labor hired from recruiters at wage w_t^{int} and f_e is the sunk entry cost in labor units, equivalent to $w_t^{int} f_e / z_t = \rho(N_t) f_e / \mu(N_t)$ units of the consumption basket. Firms enter until the post-dividend firm value equals the sunk entry cost:

$$\nu_t^f = \frac{\rho(N_t) f_e}{\mu(N_t)} \tag{21}$$

Combining the free entry condition (21) with the cutoff condition (20) yields:

$$\chi_t^c = R_t^f + \frac{\rho(N_t) f_e}{\mu(N_t)}$$

The cutoff rises with current profitability R_t^f and the value of a new product line $\rho(N_t) f_e / \mu(N_t)$: when conditions improve, marginal firms that would otherwise exit find it worthwhile to continue.

3.6. Aggregation

Labor market clearing implies $L_t = 1 - u_t$. Define gross consumption output $Y_t^c \equiv \rho(N_t) N_t y_t$ and retail employment $L_t^c \equiv N_t l_t^c$. Aggregate continuation costs paid by surviving firms are

$$X_t^c \equiv N_t \, E(\chi \,|\, \chi \leq \chi_t^c) = N_t p_0 \psi_c \chi_t^c \tag{22}$$

where $\psi_c \equiv \psi/(\psi+1)$. Aggregate retail profits gross of continuation costs are $\Pi_t^f = Y_t^c/\varepsilon(N_t) - X_t^c$, and recruiter profits are $\Pi_t^{int} = (w_t^{int} - w_t)L_t$. The average per-firm dividend is $d_t^f = \Pi_t^f/N_t$.

Total vacancy creation costs are the sum of sunk posting costs and fixed matching costs:

$$X_t = X_v + X_f = e_t \frac{\xi}{\xi+1} Q_t + \kappa q(\theta_t) v_t \tag{23}$$

where $X_v = x_m^{-\xi} \frac{\xi}{\xi+1} Q_t^{\xi+1}$ follows from integrating $xg(x)$ over $[0, Q_t]$. Given e_t and Q_t, the parameter x_m has no additional effect on entry costs provided $Q_t < x_m$.

Aggregating (9) with asset market clearing $a_{t+1} = a_t$ and $T_t = bu_t$, and using $Y_t^{Gross} = Y_t^c + \nu_t^f N_t^e$, real GDP net of intermediate inputs is:

$$Y_t \equiv Y_t^{Gross} - X_t - X_t^c = C_t + \nu_t^f N_t^e = w_t L_t + \Pi_t^f + \Pi_t^{int}$$

GDP comprises consumption and investment in new product lines. Moreover, decomposing output into income confirms the national accounts identity.

Using free entry (21) and $d_t^f = \Pi_t^f/N_t$, the firm-level Euler equation (11) becomes the business formation equation:

$$f_e \rho(N_t) = (1-\delta_t)\mathbb{E}_t m_{t+1} \Lambda_{t+1} \frac{\mu(N_t)}{\mu(N_{t+1})} \left\{ f_e \rho(N_{t+1}) + \frac{Y_{t+1}^c}{N_{t+1}} \left(\mu(N_{t+1}) - 1 - \mu(N_{t+1}) \frac{X_{t+1}^c}{Y_{t+1}^c} \right) \right\}$$

The left-hand side is the current cost of creating a product line. The right-hand side is the expected discounted return, adjusted for survival probability $(1-\delta_t)\Lambda_{t+1}$ and the current-to-future markup ratio $\mu(N_t)/\mu(N_{t+1})$. The term in braces is the next-period firm value plus the private marginal product of a new product line net of continuation costs; the net markup $\mu(N_{t+1}) - 1$ governs the return to entry.

3.7. Definition of equilibrium

Definition 1. *Given $\rho(N)$, $\mu(N)$, F_χ, initial exogenous states $\{z_0, \delta_0, s_0\}$, and initial endogenous states $\{v_{-1}, u_{-1}, u_0, N_0\}$, an equilibrium is a sequence $\{N_t, C_t, \chi_t^c, Y_t^c, \theta_t, K_t, Q_t, e_t, v_t, u_t\}_{t=0}^{\infty}$ satisfying the business formation block:*

$$N_t = (1-\delta_{t-1})\Lambda_t \left(N_{t-1} + \frac{z_{t-1}(1-u_{t-1})}{f_e} - \frac{Y_{t-1}^c}{f_e \rho(N_{t-1})} \right) \tag{24}$$

$$f_e \rho(N_t) = (1-\delta_t)\mathbb{E}_t m_{t+1} \Lambda_{t+1} \frac{\mu(N_t)}{\mu(N_{t+1})} \left\{ f_e \rho(N_{t+1}) + \frac{Y_{t+1}^c}{N_{t+1}} \left(\mu(N_{t+1}) - 1 - \mu(N_{t+1}) \frac{X_{t+1}^c}{Y_{t+1}^c} \right) \right\} \tag{25}$$

$$Y_t^c = C_t + X_t + X_t^c \tag{26}$$

$$\chi_t^c = \frac{Y_t^c(\mu(N_t)-1)}{\mu(N_t)N_t} + \frac{f_e\rho(N_t)}{\mu(N_t)} \tag{27}$$

and the labor market block:

$$\kappa + \frac{K_t}{q(\theta_t)} = \mathbb{E}_t m_{t+1}(1-\delta_t)\Lambda_{t+1}\left((1-\phi)\left(\frac{\rho(N_{t+1})z_{t+1}}{\mu(N_{t+1})} - K_{t+1} - b\right)\right.$$
$$\left. - \phi\theta_{t+1}\left(K_{t+1} + q(\theta_{t+1})\kappa\right) + (1-s_{t+1})\left(\kappa + \frac{K_{t+1}}{q(\theta_{t+1})}\right)\right) \tag{28}$$

$$K_t = Q_t - (1-\delta_t)\mathbb{E}_t m_{t+1}\Lambda_{t+1}Q_{t+1} \tag{29}$$

$$e_t = \left(\frac{Q_t}{x_m}\right)^{\xi} \tag{30}$$

together with laws of motion (6)–(5), auxiliary definitions $\delta_{e,t}$ (3), τ_t (4), Λ_t (18), X_t (23), X_t^c (22), m_t (10), and stochastic processes (7)–(8).

The business formation block jointly determines N_t, C_t, Y_t^c, and χ_t^c: the firm LOM propagates the product-line stock, the business formation Euler governs entry, the resource constraint clears goods markets, and the cutoff equates the exit threshold to current profitability plus the option value of a new product line. The labor market block jointly determines u_t, v_t, θ_t, K_t, and Q_t: the job creation condition equates average hiring cost to the present discounted value of a filled vacancy, and free entry pins down the vacancy creation margin. The two blocks interact through $\rho(N_t)$ and $\mu(N_t)$ in the job creation condition and through δ_t in the survival-adjusted discount factor $(1-\delta_t)\Lambda_{t+1}$, which appears in both.

3.8. Limiting cases

The next several propositions characterize the relationship between the business formation block and labor market block, and establish the connection to benchmark frameworks in the literature. Proofs are in Appendix Appendix C.

Proposition 1 (Independence of labor market block)**.** *Consider general CES preferences with elasticity of substitution ε held fixed. The labor market block $\{u_{t+1}, v_t, e_t, Q_t, K_t, w_t\}_{t=0}^{\infty}$ can be solved independently of the business formation block if and only if $\sigma = 0$, $\zeta = 0$, and $p_0 = 0$.*

When $\sigma = 0$, risk neutrality fixes $m_{t+1} = \beta$, eliminating the consumption dependence of the stochastic discount factor. When $\zeta = 0$, the taste for variety vanishes and $\rho(N_t) = 1$, removing N_t from recruiter compensation $w_t^{int} = \rho(N_t)z_t/\mu(N_t)$ and from the job creation condition while leaving μ and hence the business formation block otherwise intact. When $p_0 = 0$, endogenous exit is absent so $\Lambda_t = 1$ and $\delta_{e,t} = \delta_{t-1}$ depends only on the exogenous shock. Under all three conditions, the labor market block closes on its own; the business formation block then determines C_t and N_t taking u_t and X_t as given from the labor market block.

The model nests GS and CK as the taste for variety vanishes, which under DS-CES corresponds to $\varepsilon \to \infty$. This limit is not directly workable, however: with f_e fixed, vanishing profits ($\mu \to 1$) make entry unprofitable and $N \to 0$ from the resource constraint, so that $Y_t^c = \rho(N_t)N_t y_t \to 0$ with positive employment, rendering aggregate production degenerate. One might restore a non-degenerate limit by setting $\delta \to 0$, but this would eliminate the destruction margin that is central to the vacancy law of motion in GS and CK — precisely the channel we wish to preserve. We instead seek a joint limit in which entry costs shrink proportionally with profits, keeping the product-line stock well-defined and positive while making business formation irrelevant to aggregate labor market allocations. Proposition 2 shows that scaling $f_e = \bar{f}_e/\varepsilon$ jointly with $\varepsilon \to \infty$ achieves exactly this.

Proposition 2 (Double limit)**.** *Consider DS-CES preferences. Set $p_0 = 0$, and let $\varepsilon \to \infty$ jointly with $f_e = \bar{f}_e/\varepsilon$ for fixed $\bar{f}_e > 0$. Then:* (i) *the product-line mass has a unique positive steady-state value $\bar{N} > 0$ satisfying $\bar{\delta}\bar{N} = \bar{z}\bar{L}\ln \bar{N}/\bar{f}_e$; local stability of the N_t dynamics around $\bar{N}$ holds provided $\bar{N} > e^{1-\bar{\delta}}$, a condition satisfied for sufficiently small $\bar{f}_e$ given other parameters.* (ii) *firm value ν_t^f, investment $\nu_t^f N_t^e$, and labor devoted to entry l_t^e all vanish at rate $1/\varepsilon$;* (iii) *recruiter compensation $w_t^{int} \to z_t$; and* (iv) *aggregate allocations $\{\theta_t, u_t, v_t, C_t, Y_t^c\}$ are independent of $\bar{N}$, so the normalization $\bar{N} = 1$ is without loss of generality for aggregate allocations. In the limit, the business formation block is irrelevant to real allocations.*

As $\varepsilon \to \infty$ under DS-CES, $\zeta = 1/(\varepsilon - 1) \to 0$, satisfying one of the separability conditions of Proposition 1. Proposition 2 is stronger, however: it establishes not merely that the labor market block separates but that business formation becomes irrelevant to aggregate allocations entirely — a consequence of profits and entry costs vanishing jointly at the same rate. Proposition 3 below implements the limiting economy directly via clone replacement, maintaining $N_t = 1$ by construction rather than through the limit passage.

Proposition 3 (Clone-replacement equilibrium)**.** *Under clone replacement with $N_t = 1$ and $\mu = 1$, the business formation block drops out and equilibrium labor market variables $\{u_{t+1}, v_t, e_t, Q_t, K_t\}_{t=0}^{\infty}$*

satisfy:

$$
\begin{aligned}
\kappa + \frac{K_t}{q(\theta_t)} &= \mathbb{E}_t m_{t+1}(1-\delta_t)\left((1-\phi)(z_{t+1}-K_{t+1}-b)\right. \\
&\qquad \left. -\phi\theta_{t+1}(K_{t+1}+q(\theta_{t+1})\kappa) + (1-s_{t+1})\left(\kappa + \frac{K_{t+1}}{q(\theta_{t+1})}\right)\right) \qquad (31) \\
K_t &= Q_t - (1-\delta_t)\mathbb{E}_t m_{t+1} Q_{t+1} \\
v_t &= (1-\delta_{t-1})\left[(1-q(\theta_{t-1}))v_{t-1} + s_{t-1}(1-u_{t-1})\right] + e_t \\
u_t &= (1-f(\theta_{t-1}))u_{t-1} + \tau_t\left[(1-u_{t-1}) + f(\theta_{t-1})u_{t-1}\right] \qquad (32)
\end{aligned}
$$

together with $e_t = G(Q_t)$, $m_t = \beta(C_t/C_{t-1})^{-\sigma}$, $C_t = z_t(1-u_t) - X_t$, *and stochastic processes* (7)–(8), *with* $\delta_{e,t} = \delta_{t-1}$ *since endogenous exit is absent by construction.*

We dub this the *augmented Gabrovski-Silva* (AGS) model. Relative to AGS, the baseline model adds two channels affecting labor market dynamics: variety effects in the job creation condition through $\rho(N_t)/\mu(N_t)$, and endogenous exit through variation in the survival cutoff χ_t^c, as developed formally in Section **??** and evaluated quantitatively in Section 5.

Remark 1 (Nested special cases of AGS)**.** *The AGS model nests two benchmark frameworks as limiting cases. Setting* $\sigma = \kappa = 0$ *renders the AGS system isomorphic to Gabrovski and Silva (2025), extended here with aggregate AR(1) shocks to* δ_t *and* s_t*. Additionally setting* $s_t = 0$ *recovers Coles and Moghaddasi Kelishomi (2018) exactly, with* $w_t^{int} = z_t$ *and consumption determined residually from* $z_t(1-u_t) = C_t + \frac{\xi}{\xi+1}Q_t^{\xi+1}$.

3.9. Steady state

A steady state is a fixed point of the equilibrium system in which all variables are constant and shocks are at their long-run means $(\bar{z}, \bar{\delta}, \bar{s})$. We characterize it in two blocks. The *exit block* in (χ^c, δ_e) space determines the exit threshold and total destruction rate given aggregate profitability. The *labor market and entry block* in (θ, N) space determines market tightness and product variety given δ_e. The two blocks interact through δ_e and $\rho(N)$, and must be solved jointly. We focus on the (θ, N) block for the diagrammatic analysis, treating δ_e as a parameter; the full steady-state system and key derivations are in Appendix A.

Key steady-state ratios.. Setting $\delta_t = \bar{\delta}$ and $s_t = \bar{s}$, the Euler equations yield $N^e/N = \delta_e/(1-\delta_e)$. Retail profits as a fraction of consumption output are

$$\pi_s \equiv \frac{Nd^f}{Y^c} = \frac{1}{\varepsilon(N)} - \frac{X^c}{Y^c} \qquad (33)$$

The profit share π_s captures the net return to operating a product line after continuation costs: it falls with the elasticity of substitution (competition erodes markups) and with the continuation cost burden X^c/Y^c. The equilibrium relationship between total employment and product lines is:

$$N = \frac{\pi_s z L(1-\delta_e)}{f_e\left(\frac{r+\delta_e}{\mu(N)} + \delta_e \pi_s\right)} \tag{34}$$

At a given profit share, a higher destruction rate δ_e reduces product lines relative to employment by raising the effective discount rate on future profits and increasing the entry required to maintain the product-line stock. The full derivation of steady-state output shares and labor allocation is in Appendix A.

Four steady-state curves.. The *exit threshold curve* gives the locus (χ^c, δ_e) consistent with the cutoff condition (27) at steady state:

$$\chi^c = \frac{\rho(N) f_e}{\mu(N)}\left(\frac{r+\delta_e}{1-\delta_e} \cdot \frac{r+\delta_e+\psi_c(1-\delta_e)}{(1-\psi_c)(r+\delta_e)} + 1\right) \tag{35}$$

A higher destruction rate requires higher per-period profits to compensate, raising χ^c; higher variety N shifts the curve upward through $\rho(N)$. The *product destruction curve* follows from $\Lambda(\chi^c)(1-\delta) = (1-\delta_e)$:

$$\chi^c = \left(\frac{1-\delta_e}{1-\delta}\right)^{1/\psi} \chi_m \tag{36}$$

It is negatively sloped: a higher cutoff χ^c raises the endogenous survival probability and therefore lowers δ_e.

For the labor market block, express e and K as functions of tightness θ using the steady-state unemployment and vacancy laws of motion:

$$e(\theta) = \delta_e \frac{\tau\theta + (1-\delta_e) f(\theta)}{\tau + (1-\delta_e) f(\theta)}, \qquad K(\theta) = \frac{r+\delta_e}{1+r}\, e(\theta)^{1/\xi}\, x_m$$

The *job creation curve* is the steady-state JCC (28):

$$\left(\kappa + \frac{K(\theta)}{q(\theta)}\right)(r+\tau+(1-\delta_e)\phi\theta q(\theta)) = (1-\delta_e)(1-\phi)\left(\frac{\rho(N)z}{\mu(N)} - K(\theta) - b\right) \tag{37}$$

Its right-hand side is increasing in N through $\rho(N)$: more product lines raise recruiter compensation, stimulating vacancy creation — the employment–variety feedback emphasized by Schaal and Taschereau-Dumouchel (2019) and derived here in a more flexible setting. Under general

CES with $\zeta = 0$, $\rho = 1$ and (37) determines θ independently of N, yielding a vertical line in (θ, N) space. The *resource constraint curve* combines the aggregate resource constraint, free entry (21), and the N-L relationship (34):

$$N = \frac{(\mu(N) - 1)zL(\theta)(1 - \delta_e)}{f_e(\delta_e \mu(N) + r)} \tag{38}$$

Employment $L(\theta)$ is increasing and concave in θ, so (38) is increasing, concave, and bounded above by $\bar{N} = (\mu - 1)z(1 - \delta_e)/(f_e(\delta_e \mu + r))$, the product variety consistent with full employment.

Proposition 4 (Variety effects and elasticity of substitution)**.** *The exit threshold curve* (35)*, job creation curve* (37)*, and resource constraint curve* (38) *each depend on the elasticity of substitution* $\varepsilon(N)$*. Only the exit threshold and job creation curves depend on the benefit of variety* $\zeta(N)$*. The product destruction curve* (36) *is independent of both.*

The result follows directly from the expressions above: $\varepsilon(N)$ enters $\mu(N)$ and π_s in all three non-destruction curves, while $\zeta(N)$ enters only through $\rho(N)$, which is absent from the resource constraint curve (38). The product destruction curve depends only on ψ, χ_m, and $\bar{\delta}$. The implication is that empirical variation in exit rates across recessions identifies the destruction curve independently of preference parameters.

Proposition 5 (Steady-state equilibria)**.** *Letting* $p_0 \to 0$*, so that* δ_e *collapses to a parameter, there generically exist two steady-state equilibria in* (θ, N) *given by* (37) *and* (38)*. One equilibrium features strictly lower market tightness and fewer product lines. Multiple equilibria arise whenever* $\lim_{\theta \to 0} N(\theta) > 0$ *from* (37)*, which holds whenever* $b > 0$ *or* $\kappa > 0$*: in the limit* $\theta \to 0$*,* $e(\theta) \to 0$ *and* $K(\theta) \to 0$*, so* (37) *implies*

$$N^0_{JC} \equiv \lim_{\theta \to 0} N_{JC}(\theta) = \left(\frac{\mu}{z} \left(\frac{\kappa(r + \tau)}{(1 - \delta_e)(1 - \phi)} + b \right) \right)^{\varepsilon - 1} > 0 \tag{39}$$

while (38) *gives* $N \to 0$ *as* $\theta \to 0$*. As* $\theta \to \infty$*,* (37) *implies* $N \to \infty$ *while* (38) *implies* $N \to \bar{N} < \infty$*. The two curves therefore cross exactly twice, as depicted in Figure 4. Under DS-CES with* $\varepsilon \to \infty$*, or under general CES with* $\zeta \to 0$*, variety effects vanish from* (37)*, yielding a unique equilibrium with* θ *determined independently. We focus on the equilibrium with higher* (θ, N) *throughout.* Proof in Appendix A.

Figure 4 illustrates the two equilibria. A positive productivity shock z shifts both curves outward: higher z raises recruiter compensation $w^{int} = \rho(N)z/\mu$, increasing the surplus from

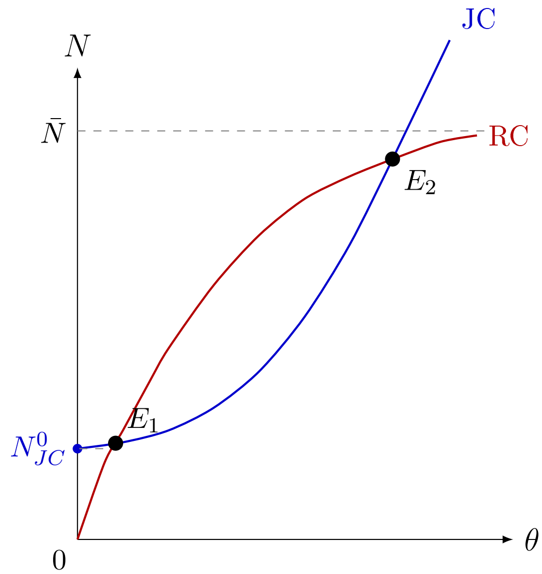

Figure 4: Steady-state equilibria in (θ, N) space. The job creation curve (JC) is convex and intersects the vertical axis at $N^0_{JC} > 0$ whenever $b > 0$ or $\kappa > 0$; see (39). The resource constraint curve (RC) is concave, starts at the origin, and is bounded above by $\bar{N} = (\mu - 1)\bar{z}(1 - \bar{\delta}_e)/(f_e(\bar{\delta}_e\mu + r))$. Two equilibria exist generically; E_1 is the low and E_2 the high tightness–variety equilibrium.

posting vacancies and shifting (37) rightward; simultaneously higher aggregate income supports more product lines for a given employment level, shifting (38) upward. The high equilibrium moves from E_2 to E_2' with both higher tightness and more product lines, as illustrated in Figure 5.

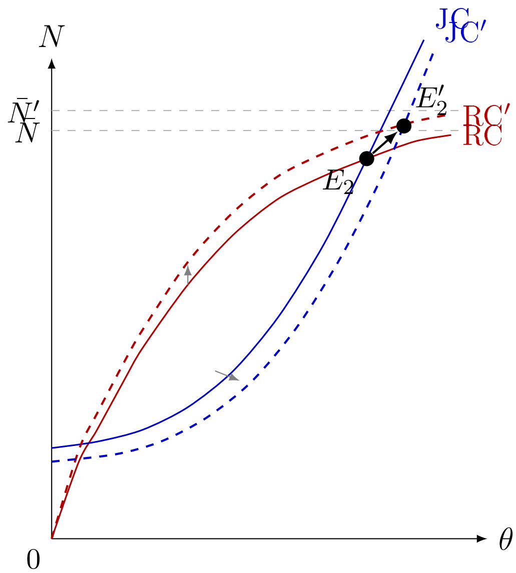

Figure 5: Effect of a positive productivity shock $z \uparrow$ on the steady-state equilibrium. Higher z raises recruiter compensation $w^{int} = \rho(N)z/\mu$, shifting the job creation curve rightward (JC to JC′); simultaneously higher income supports more product lines for given employment, shifting the resource constraint curve upward (RC to RC′, with $\bar{N}' > \bar{N}$). The high equilibrium moves from E_2 to E_2', with strictly higher market tightness and product variety.

Proposition 6 (Comparative statics)**.** *Restricting attention to the high-(θ, N) equilibrium, market tightness and product variety are jointly higher under: higher productivity z, stronger taste for variety ζ, lower worker bargaining power ϕ, lower unemployment benefits b, lower fixed entry costs f_e, lower exogenous destruction rate $\bar{\delta}$, lower elasticity of substitution ε, lower rate of time preference r, and lower fixed matching cost κ. Lower ε and lower separation rate $\bar{s}$ increase steady-state product variety N but have ambiguous effects on market tightness θ.*

4. Empirical Evidence and Bartik Local Projections

4.1. Overview and identification strategy

The cross-state evidence in Section 1 establishes that states with structurally higher establishment exit rates accumulate larger unemployment gaps during recessions, but the correlation conflates the causal effect of exit shocks with persistent differences in industrial composition, institutional environment, and average firm age across states. A state panel regression of labor market outcomes on local exit rates would confound the structural δ shock with the local demand conditions that drive both exits and unemployment simultaneously — the classic omitted variable in any cross-sectional or within-state identification strategy. The standard panel remedy of adding state and time fixed effects absorbs permanent cross-state heterogeneity and aggregate time variation but leaves untouched the within-state, within-time comovement between local exit activity and local business conditions.

We address this using a Bartik shift-share design (Bartik, 1991; Goldsmith-Pinkham, Sorkin, and Swift, 2020). The instrument for state s at quarter t is a weighted average of *national* industry-level establishment exit rates, where the weights are the state's pre-determined industry employment shares. To see why this breaks the confound, consider Michigan in 2008: its instrument is large because manufacturing nationally — not just in Michigan — was experiencing a wave of permanent closings, and Michigan had a high pre-recession exposure to manufacturing. The national exit rate is computed leaving out Michigan's own establishments (leave-one-out), so the instrument cannot be mechanically driven by Michigan-specific conditions. The instrument thus isolates variation in exit exposure driven by a state's inherited industry structure interacting with national industry cycles, rather than by local demand or policy.

The identifying assumption is that, conditional on state and time fixed effects and lagged controls, the Bartik instrument is uncorrelated with state-specific shocks to unemployment and vacancy creation beyond the exit channel. Section 4.2 describes instrument construction in detail; Section 4.3 states the local projection specification; Section 4.4 reports results. Section 4.4 provides two validation exercises: a severity placebo that tests whether the vacancy IRF is driven by recession depth rather than exit exposure, and a joint local projection that checks robustness to controlling simultaneously for the layoff-and-discharge instrument.

4.2. Instrument construction

BED data and the permanence problem.. The underlying data source is the BLS Business Employment Dynamics (BED) survey, which tracks establishment-level employment via state unemployment insurance payroll records at quarterly frequency from 1992Q3. An establishment is classified as a "closing" in a given quarter if it exits the UI payroll base between consecutive March reference dates. Coverage is all private-sector establishments across 12 supersectors and all 50 states.

Raw BED closings conflate two economically distinct events: *permanent exits* — establishments that dissolve irreversibly, destroying the associated product line and vacancy slot — and *temporary shutdowns*, establishments that suspend operations for one to three quarters before reopening. The model's δ shock corresponds exclusively to permanent exit: only an irreversible dissolution removes a product line from the variety space and eliminates the recruiter's vacancy. Using raw closings as the shock proxy therefore overstates the true destruction margin and introduces attenuation bias when the temporary-shutdown share varies over time. The contamination is most severe during the 2020Q2 lockdown, where roughly 400,000 of the recorded closings reflected government-mandated suspensions that subsequently reopened, but it is present at lower intensity throughout the sample: seasonal businesses, firms temporarily suspending operations during downturns, and establishments undergoing ownership transitions all appear as closings in BED without constituting permanent product-line destruction.

BED Deaths as the measure of permanent exits.. We address this using a second BED data series that resolves the permanence problem directly. In addition to closings, BED separately reports *deaths* — establishments absent from the UI payroll base for four or more consecutive quarters — which by construction excludes temporary shutdowns.[12] Since a temporarily closed establishment re-enters the payroll base within one to three quarters, it never accumulates four consecutive absences and therefore never appears in BED Deaths. Our δ instrument uses BED Deaths — not raw closings — as the numerator of the national exit rate.

Figure 6 shows employment-weighted mean BED Closings and Deaths rates across 12 supersectors from 1992Q3 to 2019Q4, together with 25th–75th industry percentile bands. The Deaths/Closings ratio (lower panel) averages 0.63 across the full sample and is remarkably stable over the business cycle, dipping modestly during the GFC but remaining well below unity throughout. This stability implies that the fraction of closings that are temporary does not vary sharply over time, and that using Deaths directly gives a cleaner series than either raw closings or a time-varying permanence correction.

The Deaths/Closings ratio exhibits substantial heterogeneity across sectors. Construction has the lowest mean ratio (0.57), consistent with the prevalence of project-based establishments that close on contract completion and reconstitute under a new entity — a genuinely temporary exit from BED's perspective. Leisure and hospitality is similarly low (0.82), reflecting seasonal closures. At the other end, manufacturing (1.00 at the cap), information (1.00), and finance and

[12] BED data classes: 06 (Closings) records establishments absent from the next quarterly payroll file; 08 (Deaths) records establishments absent for four or more consecutive quarters. Deaths $\subseteq$ Closings in every industry$\times$quarter cell, so ratio is bounded above by unity by construction.

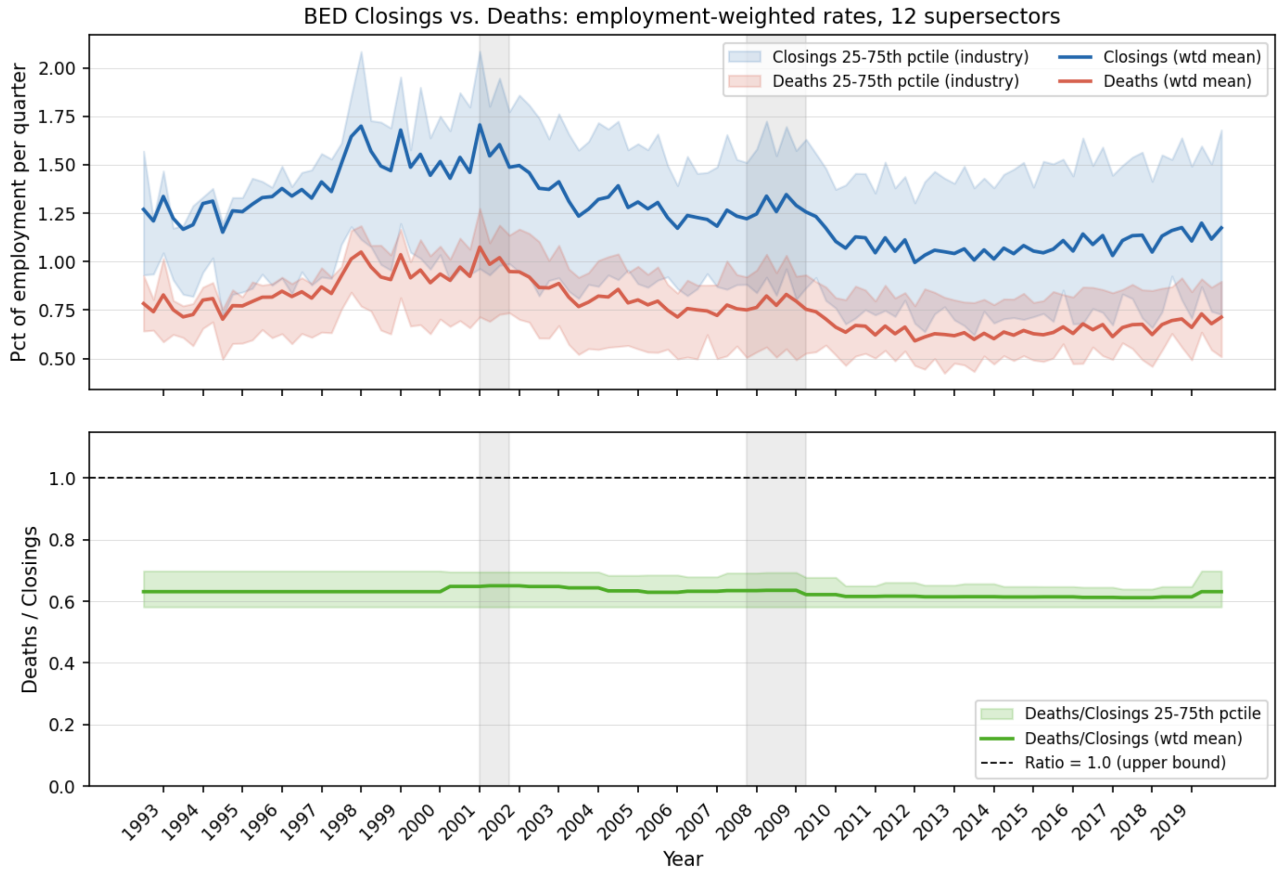

Figure 6: BED Closings and Deaths: employment-weighted rates across 12 supersectors, 1992Q3–2019Q4. *Top panel:* mean closing rate (blue) and death rate (red) as percent of employment per quarter, with 25th–75th industry percentile bands. Grey shading denotes NBER recessions. *Bottom panel:* Deaths/Closings ratio; dashed line at 1.0 is the theoretical upper bound.

real estate (1.00) hit the upper bound, implying that essentially all recorded closings in those sectors constitute permanent exits.[13] Wholesale trade (0.97), retail (0.98), and professional

[13] The Deaths/Closings ratio exceeds unity for manufacturing and information in certain years when computed from BDS exit counts rather than BED Deaths directly. The BDS-implied ratio can exceed one due to a timing mismatch: BDS records employment at the March reference date *prior* to exit while BED records employment at the time of the closing event, and establishments that permanently exit typically downsize before formal closure. A secondary source is industry reclassification: a manufacturer converting to distribution contributes a BDS death to its origin sector but may not appear as a BED closing. Neither issue arises with BED Deaths, which is constructed entirely within the BED survey. As a back-of-envelope verification, the employment-weighted mean Deaths/Closings ratio from BED (0.90, 2001–2019) closely matches the mean permanence ratio implied by cross-walking BDS annual exits against BED quarterly closings ($\bar{\pi} = 0.91$ after capping at unity), confirming that the two series measure the same underlying concept. Full industry-level BDS-implied permanence ratios

and business services (0.97) are near the cap, reflecting structurally high permanence in those sectors. The employment-weighted mean across all 12 supersectors is 0.90 (2001–2019 sample).

Most papers using BED data for shock identification (e.g., Elsby and Michaels, 2013) implicitly treat all closings as permanent. We are not aware of a prior paper using the BED Deaths series to align the instrument with the structural concept of irreversible product-line destruction.

Leave-one-out construction and base-year shares.. The leave-one-out (LOO) national permanent exit rate for state s, supersector j, quarter t is

$$g^{\delta}_{-s,j,t} \;=\; \frac{\text{deaths}_{j,t}}{E^{\text{nat}}_{-s,j,t_0}},$$

where $E^{\text{nat}}_{-s,j,t_0} = E^{\text{nat}}_{j,t_0} - E_{s,j,t_0}$ is base-year national employment in supersector j minus state s's own base-year employment. The LOO correction in the denominator ensures that state s's own exit activity cannot mechanically inflate its instrument value. Numerator LOO — subtracting state s's own deaths from the national count — is not implemented because BED state-level deaths by supersector are not publicly available. Employment shares $\omega_{s,j}$ are drawn from the 2006 QCEW (agglvl=54, private sector), a stable pre-GFC year that predates the bulk of the identifying variation. The Bartik instrument is:

$$B^{\delta}_{s,t} \;=\; \sum_{j} \omega_{s,j} \cdot g^{\delta}_{-s,j,t}. \tag{40}$$

Residualization.. Raw BED exit rates co-move with industry-specific demand conditions: when a sector contracts nationally, establishments exit at higher rates *and* surviving firms simultaneously reduce vacancy posting and raise layoffs. An instrument built on raw $g^{\delta}_{j,t}$ would therefore partly identify the effect of industry demand contractions rather than the structural δ shock. We address this by projecting each log shock rate on predetermined demand controls before Bartik aggregation, retaining the residual as the instrument input.

The baseline residualization (v2) runs a within-industry OLS regression:

$$\log g^{\delta}_{j,t} \;=\; \alpha_j + \gamma\,\Delta\log p_t + \lambda\,\Delta\log VA_{j,t-1} + \nu^{\delta}_{j,t}, \tag{41}$$

where $\Delta\log p_t$ is aggregate nonfarm business productivity growth (FRED: OPHNFB) and $\Delta\log VA_{j,t-1}$ is real value-added growth in supersector j lagged one quarter (BEA via FRED series RVAM,

$\pi_{j,y}$ are reported in Appendix ??.

RVAC, etc.). Both regressors are predetermined relative to the current-period shock draw: aggregate productivity is contemporaneous but common to all industries and thus absorbed by the time fixed effects in the second stage; industry VA is lagged one quarter to ensure strict predetermination. The residual $\nu^{\delta}_{j,t}$ is then Bartik-aggregated in place of the raw rate, yielding $\tilde{B}^{\delta}_{s,t}$.

The FRED BEA real value-added series begin in 2005Q1, which would restrict the δ residualization sample to 2005Q2. To recover the full 1997Q1+ sample, we extend the quarterly VA series back to 1992Q1 using a constrained Chow-Lin (1971) temporal disaggregation. An indicator regression estimated on the 2005Q1+ post-sample relates quarterly sectoral VA growth to real GDP growth: $\Delta \log VA_{j,t} = \alpha_j + \beta_j \, \Delta \log \mathrm{GDP}_t + \varepsilon_{j,t}$. The pre-2005 quarterly path is recovered by applying the implied growth rates $\hat{\alpha}_j + \hat{\beta}_j \, \Delta \log \mathrm{GDP}_t$ recursively from the 2005Q1 anchor level. The backcast is then constrained to match published BEA annual benchmarks for 1997–2004 via a proportional Denton adjustment: for each year A with annual benchmark $Y_{j,A}$, quarterly values are scaled by $Y_{j,A}/\overline{VA}^{\text{unconstrained}}_{j,A}$ so that the within-year average equals the published figure exactly. Industries with low GDP–VA correlation (R^2 below threshold) receive $\hat{\beta}_j = 0$, using only the sector-specific mean growth rate to avoid injecting noisy cyclical variation into the pre-period.

After v2 residualization the cross-state correlation between the δ and layoffs-and-discharges (LD) Bartik instruments falls from approximately 0.54 on raw rates to 0.42.[14] The persistence of this residual correlation after controlling for industry demand cycles and aggregate productivity is informative: the shared variation is not a demand artifact but most likely reflects industry-specific financial conditions that move both exit rates and layoff rates simultaneously. This motivates the joint local projection robustness check in Section 4.4.

[14] As a robustness check we further augment equation (41) with a monetary policy sensitivity term $\psi(\phi_j \times \Delta MP_t)$, where ϕ_j is log average establishment size in supersector j (Census CBP, 2000) and ΔMP_t is the quarterly change in the Wu–Xia (2016) shadow federal funds rate (v3 specification). The v3 results are reported in Appendix ??; v2 is the baseline throughout.

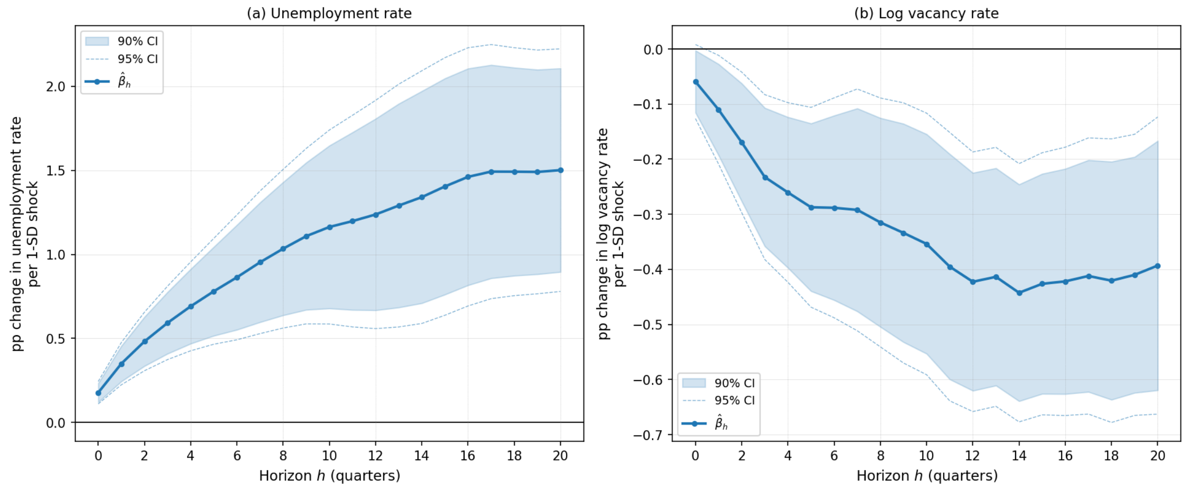

Figure 7: Impulse responses of the state unemployment rate (left) and vacancy rate (right) to a one-standard-deviation increase in the residualised δ Bartik instrument ($\hat{\sigma}_{\tilde{B}^\delta} = 14.0$ pp, 2001Q1–2019Q4). Both outcomes are estimated from panel local projections with state and time fixed effects; standard errors are clustered at the state level ($n = 50$). Shaded band: 90% confidence interval; dashed lines: 95% confidence interval. The instrument is BED establishment deaths (equation (40)), residualized on aggregate productivity growth and lagged industry value-added growth (equation (41)). Outcome window capped at 2019Q4 to exclude COVID.

4.3. Local projection specification

Unemployment outcome.. For each horizon $h = 0, 1, \ldots, 20$ quarters, the panel local projection estimating equation is

$$u_{s,t+h} - u_{s,t-1} \;=\; \alpha_s + \alpha_t + \beta_h \, \tilde{B}^\delta_{s,t} + \gamma_1 \, u_{s,t-1} + \gamma_2 \, \log L_{s,t-1} + \varepsilon_{s,t+h},$$

where $u_{s,t}$ is the state unemployment rate (LAUS), $\tilde{B}^\delta_{s,t}$ is the residualized δ Bartik instrument defined in Section 4.2, α_s is a state fixed effect, α_t is a time fixed effect, $L_{s,t-1}$ is the lagged log labor force, and $u_{s,t-1}$ is the lagged unemployment rate. The left-hand side cumulates the h-period change relative to the quarter before the shock, so β_h traces the impulse response directly. All standard errors are clustered at the state level ($n = 50$).

Vacancy outcome.. The vacancy specification is symmetric to the unemployment equation. The estimating equation at horizon h is

$$v_{s,t+h} - v_{s,t-1} \;=\; \alpha_s + \alpha_t + \beta_h \, \tilde{B}^\delta_{s,t} + \gamma_1 \, v_{s,t-1} + \gamma_2 \, \log L_{s,t-1} + \varepsilon_{s,t+h},$$

where $v_{s,t} = 100 \times V_{s,t}/L_{s,t}$ is the state vacancy rate — job openings as a percentage of the labor force — constructed from JOLTS state-level job openings (seasonally adjusted, quarterly average of monthly stocks) divided by the LAUS labor force. The left-hand side is therefore measured in percentage points, exactly as the unemployment outcome, and $\hat{\beta}_h$ has the interpretation of a percentage-point change in the vacancy rate per one-standard-deviation shock. The lagged vacancy rate $v_{s,t-1}$ is the autoregressive control; $\log L_{s,t-1}$ absorbs cross-state size differences.

Sample and standardisation.. The estimation sample runs from 2001Q1, when JOLTS state data become available, through 2019Q4; all observations with $t+h > 2019Q4$ are dropped to exclude COVID-19.[15] The instrument $\tilde{B}^{\delta}_{s,t}$ is expressed in units of percentage points of employment and has cross-sectional standard deviation $\hat{\sigma}_{\tilde{B}^{\delta}} = 14.0$ pp. All reported impulse response coefficients $\hat{\beta}_h$ are divided by $\hat{\sigma}_{\tilde{B}^{\delta}}$ before plotting, so that the scale corresponds to a one-standard-deviation shock. The sample includes all 50 states; state and time fixed effects are absorbed by within transformation before OLS estimation at each horizon.

4.4. Results

Figure 7 reports the impulse responses of the state unemployment rate and vacancy rate to a one-standard-deviation increase in the residualized δ Bartik instrument ($\hat{\sigma}_{\tilde{B}^{\delta}} = 14.0$ pp, 2001Q1–2019Q4 sample). The unemployment response rises steadily from +0.18 pp at $h = 0$ to a plateau of +1.50 pp by $h = 17$–20, remaining highly significant throughout ($p < 0.001$ from $h = 4$ onward). The vacancy response falls persistently from −0.06 pp at $h = 0$ to a trough of −0.44 pp at $h = 14$, significant at $p < 0.01$ from $h = 3$ onward and remaining so through $h = 20$. The simultaneous and persistent rise in unemployment alongside the fall in vacancies traces the outward leg of a Beveridge loop, consistent with the model's central prediction that permanent product-line destruction depletes the vacancy stock rather than merely suspending it.

A notable feature of the two IRFs is their asymmetric dynamics at medium horizons: the vacancy response stabilizes around $h = 12$–14 while the unemployment response continues to rise through $h = 20$. This pattern is consistent with a recovery path in which new vacancy creation resumes before the unemployment pool fully clears, and is further reinforced by the endogenous

[15] The δ unemployment specification can be extended back to 1997Q1 using pre-JOLTS vacancy data from the Barnichon (2010) Composite Help-Wanted Index; the extended sample attenuates the IRF modestly (peak +1.07 pp vs. +1.50 pp) because the 1997–2000 expansion years exhibit faster reabsorption of displaced workers. We use the 2001Q1 baseline throughout to maintain a common sample across unemployment and vacancy outcomes.

exit margin in the model. As the impulse to δ_t decays — with persistence $\rho_\delta \approx 0.65$, the shock is largely dissipated by $h = 6$–8 — entry of new product lines begins to restore vacancy creation, so the vacancy rate stabilizes. However, firms near the endogenous continuation threshold χ_t^c remain vulnerable to exit even after the impulse has passed, because their continuation values are still depressed by the lower product variety N_t and the compressed markup. These secondary exits extend the effective duration of the destruction episode, generating additional worker inflows into unemployment beyond what the exogenous δ persistence alone would predict. The unemployment stock therefore continues to accumulate — each secondary exit inflows more workers — while vacancies are simultaneously being replenished by new entrants. The unemployment stock drains only through matching flows, which depend on tightness $\theta = v/u$. Even as v recovers toward its steady-state value, tightness remains depressed as long as the unemployment stock is large, slowing the drain. The combination of endogenous secondary exit and the stock-flow asymmetry accounts for the divergence of the two IRFs at medium horizons. Whether the calibrated model replicates this relative timing is testable and will be assessed in Section 5.

5. Quantitative analysis

5.1. Comparison to Coles and Kelishomi

In their study, Coles and Moghaddasi Kelishomi (2018) highlight the critical role of job destruction shocks in explaining labor market fluctuations. They achieve this by incorporating finitely elastic vacancy creation and allowing for the correlation between productivity and separation shocks. Under these conditions, a job destruction shock leads to a significant increase in unemployment, even as the stock of vacancies decreases. This mechanism arises because the destruction shock not only directly reduces vacancies but also because finitely elastic vacancy creation limits the rapid increase in new vacancies despite a larger pool of unemployed workers available for matching. The oversampling of the vacancy stock by newly laid-off workers further contributes to the decline in vacancies. As a result, the job separation process accounts for a substantial portion of unemployment volatility.

The job separation shock analyzed by Coles and Moghaddasi Kelishomi (2018) affects both filled jobs and vacancies. Since each corresponds to a business opportunity, this shock is comparable to the δ shock in our model. However, the quantitative impact of a job destruction shock—and a technology shock—becomes significantly more pronounced with a higher mean value of δ. Coles and Moghaddasi Kelishomi (2018) implicitly assume that all separations inferred from unemployment flows relate to product destruction by setting $\delta = 0.031$. In contrast, data from the Business Dynamic Statistics database indicates product destruction rates in the range of $9 - 11\%$.

Further, Gabrovski and Silva (2025) demonstrate that distinguishing the match separation rate s from the destruction rate δ, and calibrating to a more moderate value, tempers the relationship between unemployment and labor productivity. As unfilled vacancies are re-posted, job creation becomes more closely linked to past productivity levels.

In our analysis, we adopt a comprehensive approach by jointly considering shocks to technology, destruction, and idiosyncratic separation. This discussion underscores the importance of the mean value of $\delta < \tau$ for transmission and motivates the inclusion of labor market volatilities, such as τ, alongside cross-correlations and autocorrelations involving τ and x. Requires updating!

5.2. Calibration and estimation via Bayesian simulated method of moments

This section sets out our estimation and calibration strategy, the taxonomy of parameters, and the explicit mapping from empirical targets to model objects.

Estimation framework. We estimate the model using a Bayesian simulated method of moments, building on Coles and Moghaddasi Kelishomi (2018), Christiano, Eichenbaum, and Trabandt (2016), and Colciago, Fasani, and Rossi (2021). The moment set extends the Shimer (2005) labor-market suite to capture volatility, co-movement, and persistence in key variables, with special attention to separations and productivity: $m(Y) = [m_1' \; m_2' \; m_3']'$, where

$$m_1 = \Big(sd(u) \quad sd(v) \quad sd(\theta) \quad sd(x) \quad sd(\tau)\Big),$$
$$m_2 = \Big(\text{Corr}(u, v) \quad \text{corr}(u, \theta) \quad \text{corr}(u, x) \quad \text{corr}(u, \tau) \quad \text{corr}(v, \theta) \quad \text{corr}(v, x) \quad \text{corr}(v, \tau)\Big),$$
$$m_3 = \Big(\text{corr}(u, u_{-1}) \quad \text{corr}(v, v_{-1}) \quad \text{corr}(\theta, \theta_{-1}) \quad \text{corr}(x, x_{-1}) \quad \text{corr}(s, s_{-1})\Big).$$

The Bayesian framework allows us to combine microeconomic priors with macroeconomic moments, quantify parameter uncertainty, and evaluate identification by comparing the posterior to the prior. For a given draw of estimated parameters, we simulate the model, compute model-implied moments, form the distance to the data, and update the posterior via MCMC. Marginal likelihoods facilitate model comparison for a fixed set of moments.

Parameter taxonomy. We partition the full parameter space Θ into three blocks.

Estimated block Θ_e: structural parameters and shock processes whose identification comes from business-cycle moments. This block contains the replacement value b; the sunk-cost share x_v of total recruiting expenditure (which maps to the fixed matching cost κ in Stage 4 below); the inverse elasticity of entry with respect to vacancy value ξ^{-1}; the endogenous exit share $\omega_\delta \equiv (1 - F(x^c))/\delta_e$; the inverse intertemporal elasticity of substitution σ; and the persistence ρ_x and conditional standard deviation σ_x of each shock $x \in \{z, \delta, s\}$.

External block: parameters set directly from microeconomic evidence or long-run data. We fix the annual discount rate $r = 4\%$ and the matching-function elasticity $\eta_L = 0.6$. Following Jaimovich and Floetotto (2008), 21% of job destruction is attributable to establishment exit; combined with an aggregate monthly separation rate of $\tau = 3.1\%$, this implies a monthly destruction rate $\delta_e = 0.651\%$ (annual: 7.54%). We set the elasticity of substitution $\varepsilon = 4.3$, implying a gross markup $\mu = \varepsilon/(\varepsilon - 1) \approx 1.30$.

Dependent block Θ_d: parameters implied jointly by the normalizations, the external block, and a draw from Θ_e via the four-stage calibration procedure described below. We normalize the steady-state mass of firms $N = 1$ and the wage $w = 1$. The remaining parameters—z, ϕ, f_e, A, s, ψ, f_m, x_m, κ, δ—are recovered from the following targets:

Target	Value	Parameter(s) identified
Annual product destruction rate δ_e^{ann}	7.54%	Monthly δ_e, exogenous rate $\delta = (1 - \omega_\delta)\delta_e$
Endogenous exit share ω_δ	estimated	Exogenous rate δ, separation rate s
Job-finding rate f^{emp}	41%	Destruction-adjusted rate f, tightness θ, level A
Vacancy-filling rate q^{emp}	80%	Destruction-adjusted rate q, tightness θ
Aggregate separation rate τ	3.1%	Match separation rate $s = (\tau - \delta_e)/(1 - \delta_e)$
Recruiting cost share X/Y	1.5%	Vacancy value Q, then κ, z, f_e, x_m, ϕ
Fixed cost share X^c/Y	10%	Within-sector ratio X^c/Y^c, profit share π_s
Wage normalization $w = 1$	—	Productivity z
Firm-mass normalization $N = 1$	—	Entry cost f_e

Calibration algorithm. Given a draw from Θ_e and the external block, we recover Θ_d in four sequential stages.

Stage 1: direct conversions (targets → fundamentals). The destruction rate target δ_e^{ann} and endogenous share ω_δ pin

$$\delta_e = 1 - (1 - \delta_e^{ann})^{1/12}, \qquad \delta = (1 - \omega_\delta)\,\delta_e.$$

The match separation rate follows from the aggregate target:

$$s = \frac{\tau - \delta_e}{1 - \delta_e}.$$

The empirical job-finding and vacancy-filling rates are corrected for destruction before use: $f = f^{emp}/(1 - \delta_e)$, $q = q^{emp}/(1 - \delta_e)$. Market tightness $\theta = f/q$, the unemployment rate $u = \tau/(\tau + (1 - \delta_e)f)$, and the matching-function level parameter $A = f/\theta^{1-\eta_L}$ are then computed analytically.

Stage 2: profit share (root-find X^c/Y^c). The fixed-cost share of retail output X^c/Y^c is not directly observed; we recover it by solving

$$\frac{X^c/Y}{(Y^c/Y^{Gross})(Y^{Gross}/Y)} = \frac{X^c}{Y^c},$$

where $Y^c/Y^{Gross} = (r + \delta_e)/(r + \delta_e + \delta_e\pi_s)$ and $Y^{Gross}/Y = 1 + X/Y + X^c/Y$. The profit share $\pi_s = 1/\varepsilon - X^c/Y^c$ is a residual: retail value added equals the Lerner markup $1/\varepsilon$ after paying stochastic fixed costs.

Stage 3: cost-distribution parameters (root-find ψ). Given π_s, we recover the mass fraction of continuing cost draws, *cons*, from

$$\pi_s = \frac{\mu - 1}{\mu}\,(1 - cons)\,\frac{r + \delta_e}{r + \delta_e + cons\,(1 - \delta_e)},$$

then map *cons* to the power-law shape parameter $\psi = (cons/p_0)\,/\,(1 - cons/p_0)$ and location parameter $f_m = x^c/\zeta^{1/\psi}$, where $\zeta = ([(1 - \delta_e)/(1 - \delta)] - (1 - p_0))/p_0$ is the quantile of the exit threshold within the continuous part of the cost distribution. Here $p_0 = 0.5$ is fixed externally as the probability that the continuation cost is positive.

Stage 4: vacancy value (root-find Q). We solve for the steady-state vacancy value Q by requiring the recruiting-cost share to equal its target: $X/Y = 1.5\%$. Given a trial value Q, the contact-rate value $K = (r+\delta_e)Q/(1+r)$ and the fixed matching cost $\kappa = (1-x_v)/x_v \cdot K/q$ follow from the sunk-cost split x_v. Total recruiting expenditure $X = e/(1+\xi^{-1})\cdot Q+\kappa qv$, the interior wage w^{int}, and firm-level output $Y^c = \rho z L^c$ are computed sequentially. The value of Q is chosen so that output implied by the recruiting-cost target, $Y = X/(X/Y)$, coincides with expenditure-side output $Y = C + \nu^f N^e$. The remaining dependent parameters are then recovered analytically:

$$z = (\mu/\rho)\, w^{int}, \tag{42}$$

$$f_e = \pi_s\, z\, L^c\, (1-\delta_e)\, \mu \,/\, (N\,(r+\delta_e)), \tag{43}$$

$$\phi = \frac{w-b}{w^{int} - K + \theta(K + q\kappa) - b}, \tag{44}$$

$$x_m = Q \,/\, e^{\xi^{-1}}.$$

The normalizations provide clean intuition: $w = 1$ pins z via (42) because the interior wage $w^{int} = \rho z/\mu$ is proportional to productivity; $N = 1$ pins f_e via (43) because free entry ties the mass of product lines to the entry cost; and the recruiting-cost share target $X/Y = 1.5\%$, once Q is found, pins bargaining power ϕ via (44).

Discussion of b and ϕ. In our labor market framework, the labor share and the search wedge w/w^{int} both depend on Nash bargaining power ϕ and the worker outside option b. Shimer (2005) interprets b as unemployment insurance and calibrates it to 0.4, finding that technology shocks generate an order-of-magnitude too little unemployment and vacancy volatility, attributing this to excessively wage-elastic labor markets. Hagedorn and Manovskii (2008) show that even adjusting ϕ to match the mild empirical wage cyclicality leaves tightness volatility unchanged; what matters is the proportional response of profits to productivity. We resolve this debate by estimating b via simulated method of moments and, given the draw of b, selecting ϕ internally to satisfy (44)—ensuring that the wage normalization and labor-share implications are mutually consistent. Our model links market power with search frictions, making the labor share a natural discipline on both ϕ and b. Matching the consumption share and labor share jointly also implies a realistic stock-market capitalization-to-output ratio without sacrificing fit on business-cycle moments.

5.3. *Posterior estimates*

5.4. *Impulse responses*

5.5. *Inspecting the mechanism*

We begin by emphasizing the significance of variety effects by comparing the impulse responses of our model with those of a similar model where $\zeta = 0$. Importantly, both versions maintain consistent markup, labor share of income, and other calibrated targets. As illustrated in Figure8, we observe that unemployment exhibits increased persistence, which is closely linked to changes in business formation and the corresponding measure of product lines. Given that the number of firms, and consequently the relative price p, is predetermined, the immediate response of unemployment remains unaffected by variety effects.

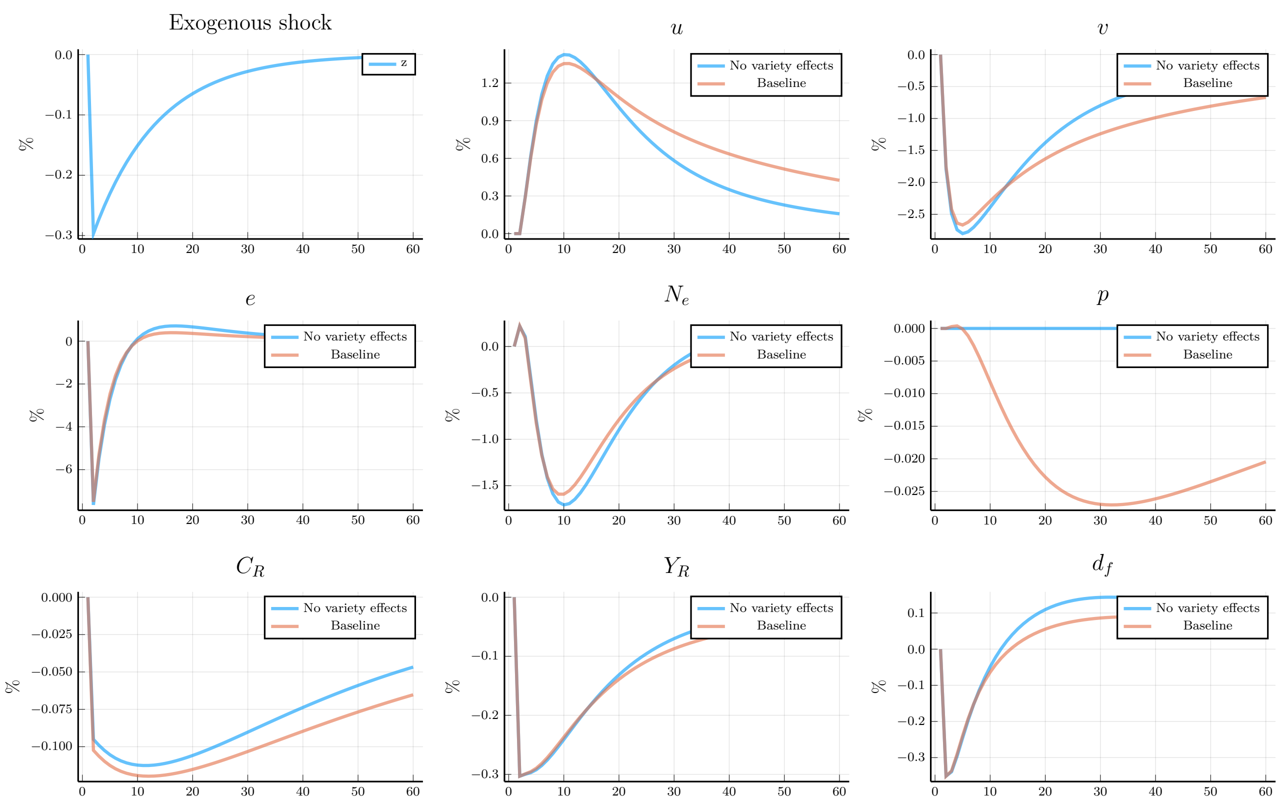

Figure 8

Second, we investigate the impact of risk aversion by comparing the effects of a destruction shock in our baseline model to those in a risk-neutral environment, typical of the benchmark search model, as depicted in Figure 9. Our primary finding is that labor market variables are generally insensitive to the consumption smoothing motive. This result arises from two factors: (1) the finite elasticity of vacancy creation inherently encourages smoothing of vacancy creation, and (2) new vacancies constitute a small portion of the overall vacancy stock. However, with either more elastic vacancy creation or increased product destruction, the responses of vacancies and unemployment would differ. Notably, risk aversion significantly influences the responses of business formation, consumption, and output. In the absence of convex adjustment costs in business formation, a persistent destruction shock substantially diminishes the present discounted value of profits under risk neutrality. Essentially, the incentive to initiate a new product line declines with a persistent destruction shock, leading households to favor scaling back business formation and increasing consumption instead. In a risk-neutral scenario, a destruction shock would prompt consumption and output to rise alongside unemployment, thereby limiting their role in driving business cycle dynamics. Therefore, when considering the dynamics of business formation—key macroeconomic aggregates—the degree of risk aversion becomes crucial.

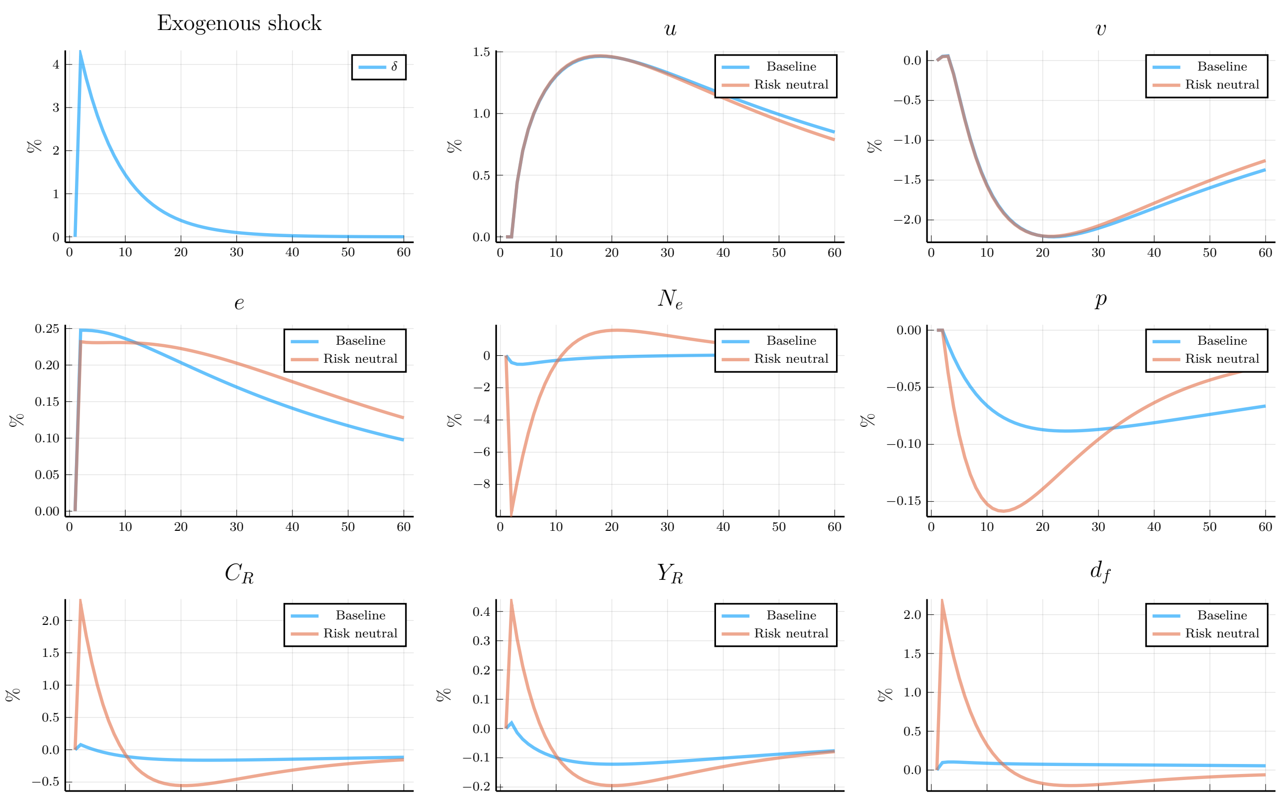

Figure 9

Third, we explore the impact of the inverse elasticity of vacancy creation ξ^{-1}. Specifically, we compare the baseline response to the model recalibrated with $\xi^{-1} = 0.5$. In this analysis, we focus on the responses to both technology and destruction shocks, noting significant differences between the two scenarios.

Figure 10 illustrates the responses under a destruction shock. With increased elasticity in vacancy creation, the number of new vacancies rises, which mitigates the increase in unemployment. This effect also leads to a smaller decline in business formation and output. Importantly, despite the heightened elasticity in entry, the negative correlation between unemployment and vacancies persists, maintaining the integrity of the Beveridge curve.

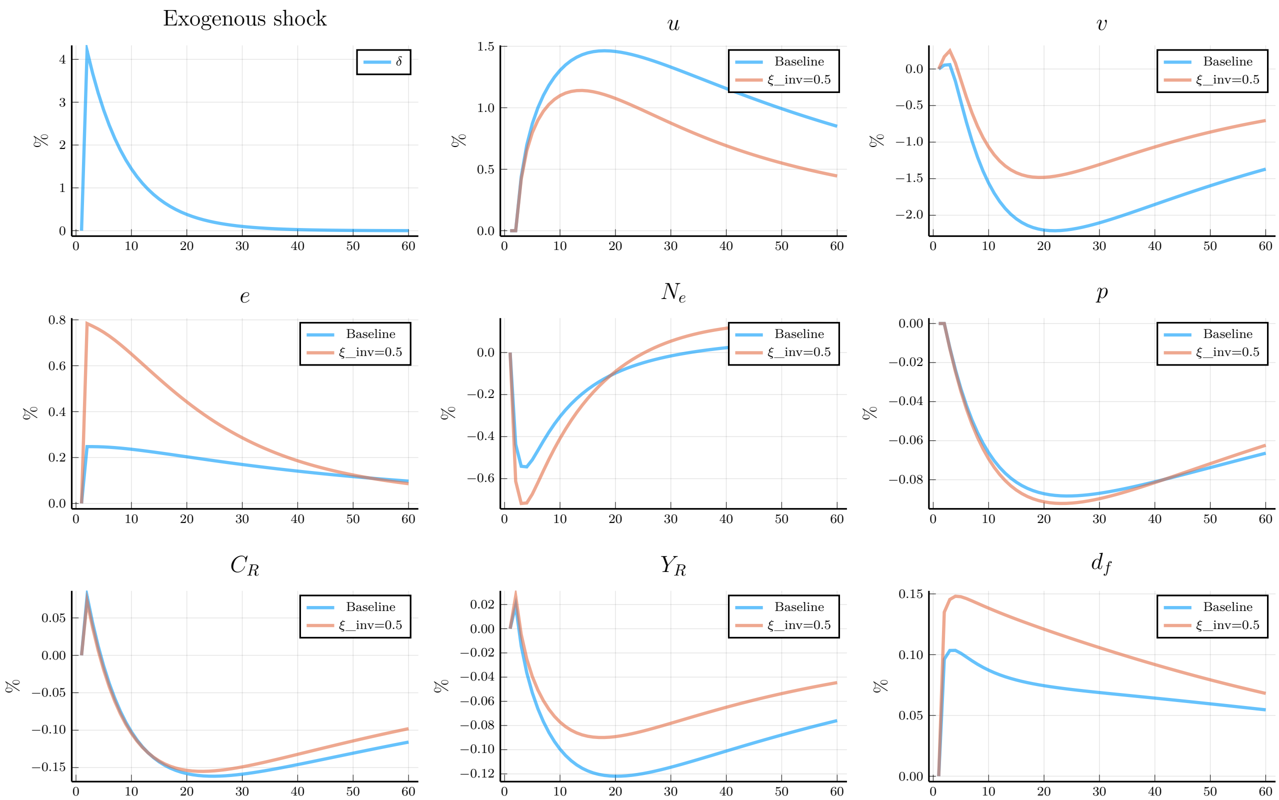

Figure 10

Figure 11 depicts the responses to a contractionary technology shock. With more elastic vacancy creation, the number of new vacancies declines significantly, leading to a substantial reduction in the overall vacancy stock and, consequently, a pronounced increase in unemployment. However, the responses of business formation, consumption, and output remain relatively unchanged. Thus, increased elasticity in vacancy creation amplifies the impact of technology shocks while diminishing the effects of destruction shocks. In other words, incorporating finitely elastic vacancy creation into the benchmark search model allows destruction shocks to have a more pronounced effect on unemployment, while reducing the influence of technology shocks.

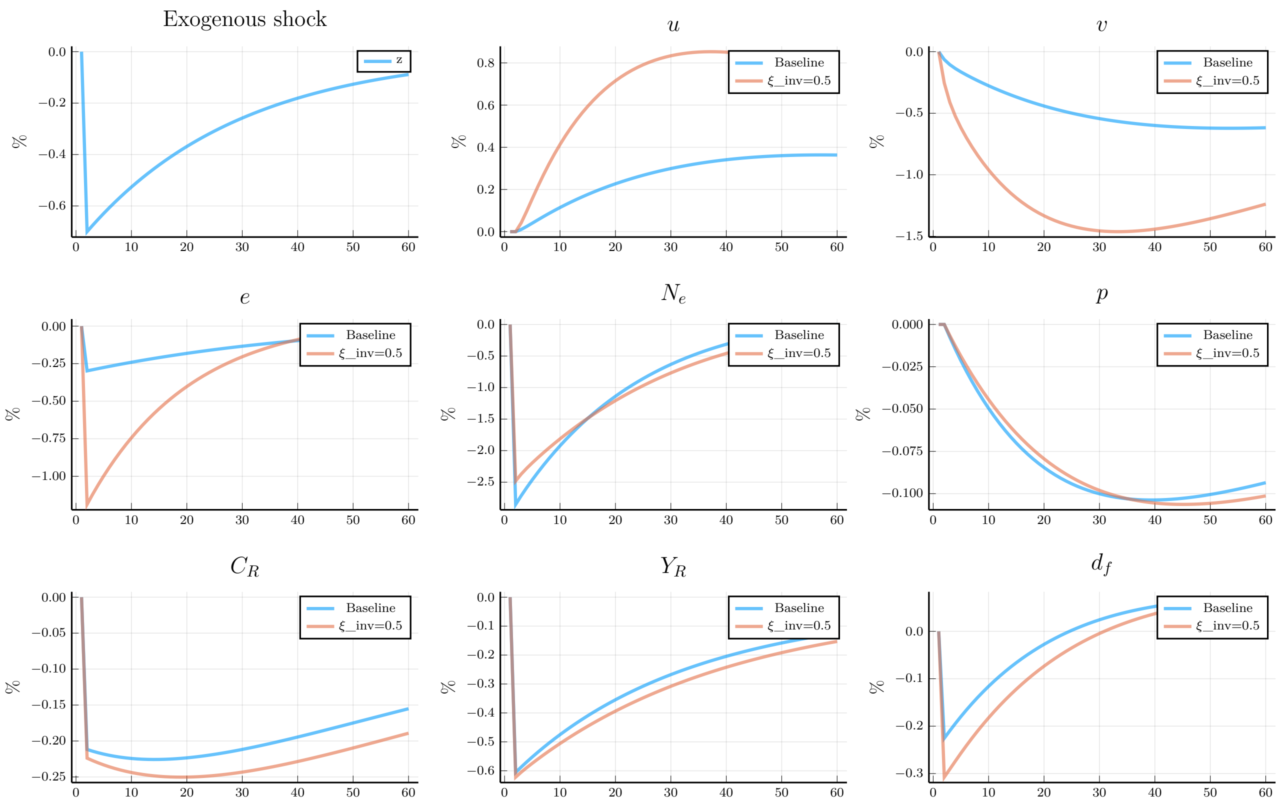

Figure 11

Pissarides (2009) proposes fixed matching costs as a means to make average hiring costs less cyclical, so that job creation becomes more sensitive to fluctuations in technology. This logic applies in our model as well, so we omit the impulse responses. However, Figure 12 shows that the responses are nearly identical in the case of a destruction shock.

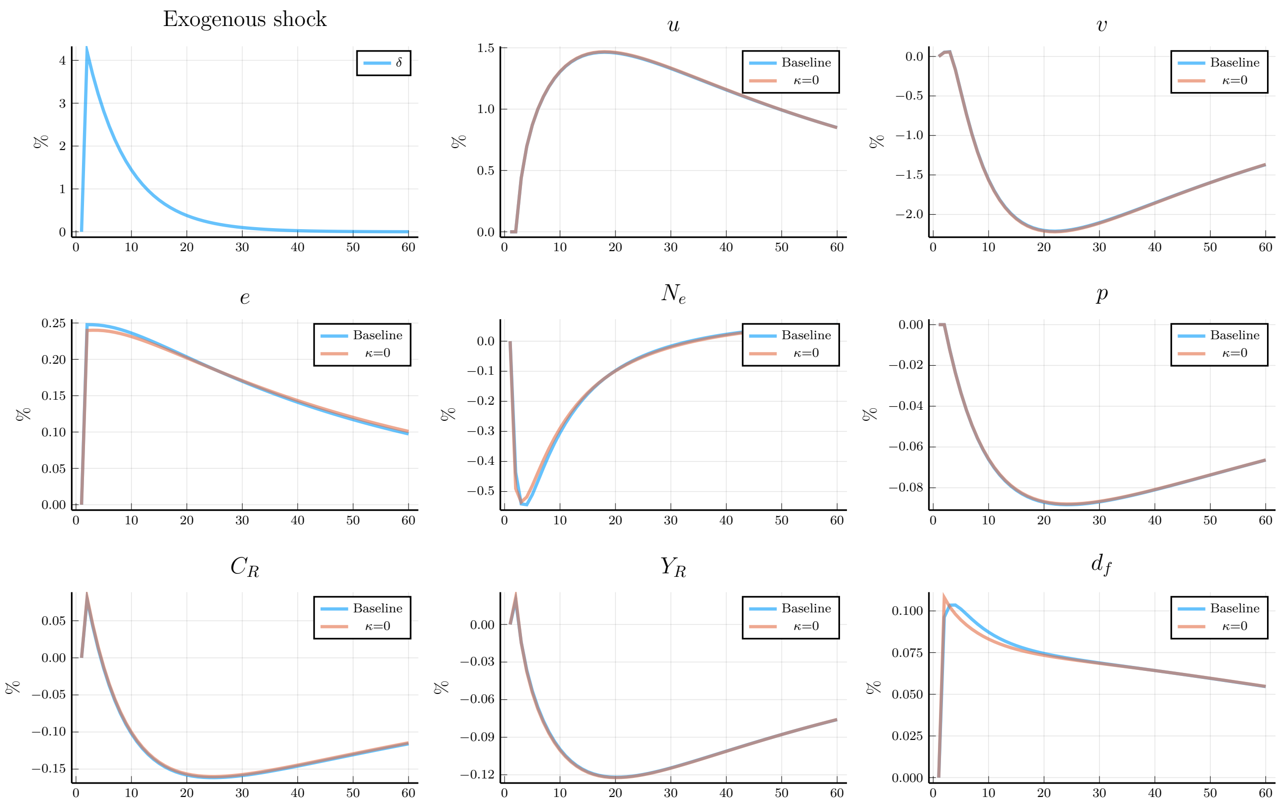

Figure 12

To do: include plot for corresponding BGM model Finally, we compare a technology shock in this framework to Bilbiie, Ghironi, and Melitz (2012). There are several key differences. First, labor reponds only via the extensive margin. It is predetermined on impact and can only expand gradually. A Frisch elasticity of labor supply of one produces a similar response of output on impact.

6. Conclusion

Appendix A. Steady state

We summarize the steady state of the system below:

$$\nu^f = \rho(N) f_e / \mu(N)$$

$$w^{int} = \rho(N) z / \mu(N)$$

$$\frac{d^f}{\nu^f} = \frac{r + \delta_e}{1 - \delta_e}$$

$$\kappa + \frac{K}{q(\theta)} = \frac{1 - \delta_e}{r + \tau}(w^{int} - w - K)$$

$$
\begin{aligned}
w &= \phi\left[w^{int} - K + \theta(K + q(\theta)\kappa)\right] + (1-\phi)b \\
Q &= e^{1/\xi}x_m \\
1-\delta_e &= (1-\delta)F(x^c), \quad \tau = 1-(1-\delta_e)(1-s) \\
K &= \frac{r+\delta_e}{1+r}Q \\
u &= \frac{\tau}{\tau + (1-\delta_e)f(\theta)} \qquad \text{(A.1)} \\
e &= \delta_e(v+1-u) \qquad \text{(A.2)} \\
N^e &= \delta_e N/(1-\delta_e) \\
L^c &= \frac{r+\delta_e}{r+\delta_e+\delta_e\pi_s\mu}L, \quad L^e = \frac{\delta_e\pi_s\mu}{r+\delta_e+\delta_e\pi_s L\mu} \qquad \text{(A.3)} \\
x^c &= \frac{\rho f_e}{\mu}\left(\frac{r+\delta_e}{1-\delta_e}\frac{\mu-1}{\mu\pi_s}+1\right) \\
\pi_s &= \frac{\mu-1}{\mu}\frac{(1-\psi_c)(r+\delta_e)}{r+\delta_e+\psi_c(1-\delta_e)} \\
Y^{Gross} &= Y^C + \nu^f N^e = w^{int}L + Nd^f \\
Y &= C + \nu^f N^e \\
X &= e\frac{\xi}{\xi+1}Q + \kappa q v \\
X^c &= N\psi_c x^c \\
Y^c &= \rho z L^c = C + X + X^c
\end{aligned}
$$

Most of these equations are self-evident from the equilibrium system. For instance, evaluating (5) yields (A.1). To find e, start with the steady-state vacancy law of motion:

$$[1-(1-\delta_e)(1-q)]v = (1-\delta_e)s(1-u) + e$$

Expanding the left-hand side as $\delta_e v + (1-\delta_e)qv$ and rewriting $(1-\delta_e)qv = \tau(1-u)$, we have

$$
\begin{aligned}
\delta_e v + \tau(1-u) &= (1-\delta_e)s(1-u) + e \\
\delta_e v + [\tau - (1-\delta_e)s](1-u) &= e
\end{aligned}
$$

so $e = \delta_e(v+1-u)$. Equation (A.2) makes clear how the contribution of new entrants to the vacancy pool is governed by the total exit rate δ_e.

To find labor in entry L^e in terms of aggregate labor, (A.3), we plug the resource constraint curve into the expression for L:

$$\begin{aligned} L^e &= \frac{\delta_e}{1-\delta_e} N f_e/z \\ &= \frac{\delta_e}{1-\delta_e}\left[\frac{(\mu-1)zL(1-\delta_e)}{f_e(\delta_e\mu+r)}\right] f_e/z \\ &= \frac{\delta_e(\mu-1)}{\delta_e\mu+r} L \end{aligned}$$

The steady-state job creation and wage equations can be arranged as

$$\kappa + \frac{K}{q} = \frac{1-\delta_e}{r+\tau}\left[(1-\phi)(w^{int}-K-b) - \phi\theta q\left(\kappa+\frac{K}{q}\right)\right] \tag{A.4}$$

Equation (A.4) equates the average recruiting cost to the present-discounted value of the match, adjusted for the wage bargain. The right-hand side incorporates a fraction $1-\phi$ of the flow surplus $(w^{int}-K-b)$ and an adjustment for the outside option of the worker. Whereas in the standard DMP model, the analogous expression would characterize tightness, here K and w^{int} are additional endogenous variables. Moreover, the average recruiting cost has the compound form $\kappa + K/q$, and discounting is adjusted by the overall firm survival rate $1-\delta_e$.

Appendix B. Derivations

Appendix B.1. Relationship between profit share and Power law shape parameter ψ

Here, we can establish the relationship between π_s and $\psi_c = \psi/(1+\psi)$. and the probability of a positive fixed cost p_0. Recall that the cutoff satisfies $x^c = Y^c[(\mu-1)/(\mu N)] + \rho f_e/\mu$. So, aggregate fixed costs are $X^c = \psi_c p_0(Y^c(\mu-1)/\mu + N\rho f_e/\mu)$. Divide by Y^c and use $Y^c = \rho z L^c$ to show that

$$\frac{X^c}{Y^c} = \frac{\psi_c p_0}{\mu}\left(\mu - 1 + \frac{f_e N}{z L^c}\right)$$

Now, plug in (**??**) to show

$$\frac{X^c}{Y^c} = \frac{\psi_c p_0}{\mu}\left(\mu - 1 + \frac{1-\delta_e}{r+\delta_e}\mu\pi_s\right)$$

Hence,

$$\pi_s = \frac{\mu-1}{\mu}(1-\psi_c p_0) - \frac{\psi_c p_0(1-\delta_e)\pi_s}{r+\delta_e}$$

We solve for π_s as

$$\pi_s = \frac{\mu - 1}{\mu} \frac{(1 - \psi_c p_0)(r + \delta_e)}{r + \delta_e + \psi_c p_0 (1 - \delta_e)} \tag{B.1}$$

It is easy to verify that, holding δ_e fixed, π_s is increasing with ψ_c. In particular, for $\psi_c = 0, \pi_s = (\mu - 1)/\mu = 1/\varepsilon$, as Bilbiie, Ghironi, and Melitz (2012). We can rewrite the cutoff x^c in terms of both the profit share and $\psi_c p_0$:

$$\begin{aligned} x^c &= \frac{\rho}{\mu} \left(\frac{z L^c (\mu - 1)}{N} + f_e \right) \\ &= \frac{\rho f_e}{\mu} \left(\frac{r + \delta_e}{1 - \delta_e} \frac{\mu - 1}{\mu \pi_s} + 1 \right) \\ &= \frac{\rho f_e}{\mu} \left(\frac{r + \delta_e}{1 - \delta_e} \frac{r + \delta_e + \psi_c p_0 (1 - \delta_e)}{(1 - \psi_c p_0)(r + \delta_e)} + 1 \right) \end{aligned}$$

Appendix B.2. Resource constraint curve

Expressing the N–L relationship in terms of the distributional parameter ψ_c, rearrange (34) as

$$N = \frac{z L (1 - \delta_e)}{f_e \left(\frac{r + \delta_e}{\mu \pi_s} + \delta_e \right)}$$

Substitute the value π_s from (B.1) to obtain

$$N = \frac{z L (\mu - 1)(1 - \psi_c p_0)}{f_e [r + \delta_e + \psi_c (1 - \delta_e) + \delta_e (\mu - 1)(1 - \psi_c)]}$$

Hence, labor in entry is

$$\begin{aligned} L^e &= \frac{\delta_e}{1 - \delta_e} \frac{N f_e}{z} \\ &= \frac{\delta_e (\mu - 1 - \psi_c p_0) L}{\delta_e \mu + r + \psi_c p_0 (1 - \mu \delta_e)} \end{aligned}$$

Thus, labor in the consumption sector is

$$L^c = \frac{r + \delta_e + \psi_c p_0 (1 - (\mu - 1)\delta_e)}{\delta_e \mu + r + \psi_c p_0 (1 - \mu \delta_e)} L$$

We can substitute to solve for X^c as

$$X^c = \frac{\rho z L}{\mu} \psi_c p_0 \left(\mu - 1 + \frac{(\mu - 1 - \psi_c p_0)(1 - \delta_e \mu)}{\delta_e \mu + r + \psi_c p_0 (1 - \mu \delta_e)} \right)$$

Hence, the ratio X^c / Y^c satisfies

$$\frac{X^c}{Y^c} = \frac{L}{L^c} \frac{\psi_c p_0}{\mu} \left(\mu - 1 + \frac{(\mu - 1 - \psi_c p_0)(1 - \delta_e \mu)}{\delta_e \mu + r + \psi_c p_0 (1 - \mu \delta_e)} \right)$$

Appendix B.3. Labor share of GDP

The aggregate resource constraint can be rearranged to pin down the income of recruiters:

$$w^{int}L = \frac{Y^c}{\mu} + \nu^f N^e$$

Thus, the income share of recruiters relative to gross output is

$$\begin{aligned}\frac{w^{int}L}{Y^{Gross}} &= \left(\frac{r+\delta_e}{r+\delta_e+\delta_e\pi_s}\right)\frac{1}{\mu} + \frac{\delta_e\pi_s}{r+\delta_e+\pi_s} \\ &= \frac{r+\delta_e+\delta_e\mu\pi_s}{\mu(r+\delta_e+\delta_e\pi_s)}\end{aligned}$$

The labor share of GDP is

$$\frac{wL}{Y} = (1+\frac{X+X^c}{Y})\frac{w}{w^{int}}\frac{r+\delta_e+\delta_e\mu\pi_s}{\mu(r+\delta_e+\delta_e\pi_s)}$$

The steady-state labor share is composed of three factors. The first factor just reflects the wedge between gross output and GDP, which depends on the recruiting cost and fixed cost shares. The second and third comprise a search wedge w/w^{int} and a market power wedge $w^{int}L/Y^{Gross}$. Through the latter, the labor share depends positively on δ_e and ρ, and negatively on π_s. Intuitively, a higher product destruction rate requires more labor resources, and thus income accruing to labor. A higher price markup raises profits and thus lowers the labor share. Indeed, in standard DSGE models with monopolistic competition, the labor share just the inverse of the price markup. Here that arises as a limiting case in which search frictions vanish ($w \to w^{int}$) and $\delta \to 0$. The latter entails eliminating the entry sector, which affects the labor share since only retailers charge markups. Note that the search wedge, in turn, depends on ϕ and b. Moreover, variety effects affect the search wedge through w^{int} but otherwise do not affect the steady-state shares.

The special case studied by Bilbiie, Ghironi, and Melitz (2012) arises if we shut down labor market frictions and fixed costs in production. In this limit $\delta_e = \delta$ (no endogenous exit), $w^{int} = w$, $X = X^c = 0$, and $\pi_s = 1/\varepsilon$, giving

$$\frac{wL}{Y} = \frac{(r+\delta)(\varepsilon-1)+\delta}{\varepsilon(r+\delta)+\delta} = \frac{C}{Y}\frac{1}{\mu} + \frac{\nu^f N^e}{C}$$

which equals the consumption share divided by the markup plus the entry share.

Appendix B.4. Wage determination

We characterize the wage w_t. To characterize these value functions, we revisit the value of the large household:

$$W_t = \max_{c_t, a_t} \frac{c_t^{1-\sigma}}{1-\sigma} + \beta \mathbb{E}_t W_{t+1} \quad \text{s.t.}$$
$$c_t + \nu_t a_{t+1} = w_t L_t + b(1-L_t) + (\nu_t + d_t) a_t + D_t^{int} - T_t$$

Note that, given the continuum of members in a household, the surplus of moving an unemployed household into employment is $\frac{\partial W_t}{\partial L_t}$. Nash bargaining solves

$$w_t = \arg\max [\frac{\partial W_t(w_t)}{\partial L_t}]^{\phi} [J_t(w_t) - Q_t]^{1-\phi} \tag{B.2}$$

For convenience, we define the match surpluses $S_T^H = \frac{\partial W_t}{\partial L_t}$. and $S_t^J = J_t - Q_t$.

From the Bellman equation,

$$\frac{\partial W_t}{\partial L_t} = c_t^{-\sigma} \frac{\partial c_t}{\partial L_t} + \beta \mathbb{E}_t [\frac{\partial W_{t+1}}{\partial L_{t+1}} \frac{\partial L_{t+1}}{\partial L_t}] \tag{B.3}$$

From the budget constraint, $\frac{\partial c_t}{\partial L_t} = w_t - b$. Moreover, given the law of motion of employment $L_{t+1} = (1-\delta_t)[f(\theta_t)(1-L_t) + (1-s_t)]L_t$, we have

$$\frac{\partial L_{t+1}}{\partial L_t} = (1-\delta_t)(1 - s_t - f(\theta_t))$$

Putting it together, we have

$$S_t^H = c_t^{-\sigma}(w_t - b) + \beta(1-\delta_t)(1 - s_t - f(\theta_t)) \mathbb{E}_t S_{t+1}^H$$

The first-order condition of the Nash product (B.2) is

$$\phi \frac{\frac{\partial S_t^H}{\partial w_t}}{S_t^H} + (1-\phi) \frac{\frac{\partial J_t}{\partial w_t}}{J_t} = 0$$

We evaluate

$$\frac{\partial S_t^H}{\partial w_t} = \frac{\partial [u'(c_t)(w_t - b)]}{\partial w_t} + \beta(1-\delta_t)(1 - s_t - f(\theta_t)) \frac{\partial}{\partial w_t} \mathbb{E}_t S_{t+1}^H$$

The term $\frac{\partial}{\partial w_t} \mathbb{E}_t S_{t+1}^H = 0$ because the next-period wage is re-bargained based on next period conditions. Thus, $\frac{\partial S_t^H}{\partial w_t} = c_t^{-\sigma}$. Recall that the firm surplus is

$$S_t^J = w_t^{int,j} - w_t - K_t + (1-s_t) \left(\kappa + \frac{K_t}{q(\theta_t)} \right)$$

Note that $\frac{\partial S_t^J}{\partial w_t} = -1$, so we obtain the sharing rule

$$S_t^H = \frac{\phi}{1-\phi} c_t^{-\sigma} S_t^J$$

From (13),

$$\mathbb{E}_t \beta(1-\delta_t) \left(\frac{c_{t+1}}{c_t}\right)^{-\sigma} S_{t+1}^J = \kappa + \frac{K_t}{q(\theta_t)}$$

Nash bargaining at period $t+1$ corresponds to (B.3) forwarded one period. Hence,

$$\beta(1-\delta_t)\mathbb{E}_t S_{t+1}^H = \frac{\phi}{1-\phi} c_t^{-\sigma} \left(\kappa + \frac{K_t}{q(\theta_t)}\right)$$

We can rewrite the household surplus as

$$S_t^H = c_t^{-\sigma}(w_t - b) + (1 - s - f(\theta_t)) \frac{\phi}{1-\phi} c_t^{-\sigma} \left(\kappa + \frac{K_t}{q(\theta_t)}\right)$$

Now, we wish to substitute out S_t^H. To do so, we combine the expressions for recruiter surplus and bargaining:

$$S_t^J = w_t^{int} - w_t - K_t + (1-s_t)\left(\kappa + \frac{K_t}{q(\theta_t)}\right)$$
$$S_t^J = \frac{1-\phi}{\phi} c_t^{\sigma} S_t^H$$

so that

$$w_t^{int} - w_t - K_t + (1-s_t)\left(\kappa + \frac{K_t}{q(\theta_t)}\right) = \frac{1-\phi}{\phi}\left(w_t - b + (1-s-f(\theta_t))\frac{\phi}{1-\phi}\left(\kappa + \frac{K_t}{q(\theta_t)}\right)\right)$$

This expression can be rearranged explicitly for the wage:

$$w_t = \phi\left(w_t^{int} - K_t + f(\theta_t)\left(\kappa + \frac{K_t}{q(\theta_t)}\right)\right) + (1-\phi)b$$

Appendix C. Limiting cases

Proof of Proposition 1

We establish the if-and-only-if by showing necessity and sufficiency of each condition separately.

Sufficiency. Set $\sigma = 0$, $\zeta = 0$, $p_0 = 0$. Under $\sigma = 0$, the stochastic discount factor reduces to $m_{t+1} = \beta$, eliminating dependence on consumption C_t from all intertemporal conditions. Under $\zeta = 0$, general CES preferences give $\rho(N_t) = 1$ for all N_t, so recruiter compensation $w_t^{int} = \rho(N_t)z_t/\mu(N_t) = z_t/\mu$ depends on N_t only through $\mu(N_t)$; but under general CES with $\zeta = 0$, μ is a constant independent of N_t, so $w_t^{int} = z_t/\mu$ is exogenous to the labor market block. Under $p_0 = 0$, $\Lambda_t = F_\chi(\chi_t^c) = 1 - p_0 = 1$ for all χ_t^c, so $\delta_{e,t} = 1 - (1 - \delta_{t-1})\Lambda_t = \delta_{t-1}$ depends only on the exogenous shock.

Under all three conditions, inspect the labor market block (28)–(30) and laws of motion (6)–(5). With $m_{t+1} = \beta$, $w_t^{int} = z_t/\mu$, and $\delta_{e,t} = \delta_{t-1}$, these equations form a closed system in $\{u_{t+1}, v_t, e_t, Q_t, K_t, w_t\}_{t=0}^{\infty}$ given exogenous states $\{z_t, \delta_t, s_t\}$ and initial conditions $\{v_{-1}, u_0\}$. No variable from the business formation block $\{N_t, C_t, Y_t^c, \chi_t^c\}$ appears. The labor market block is therefore self-contained.

Given the solution to the labor market block, the business formation block (24)–(27) determines $\{N_t, C_t, Y_t^c, \chi_t^c\}$ taking $\{u_t, X_t\}$ as given. This confirms the recursive structure.

Necessity. We show that relaxing any single condition breaks separability.

$\sigma > 0$: With $\sigma > 0$, $m_{t+1} = \beta(C_{t+1}/C_t)^{-\sigma}$ depends on C_t, which is determined by the resource constraint (26) jointly with N_t, X_t^c, and Y_t^c. The job creation condition (28) and flow vacancy value (29) both contain m_{t+1}, so the labor market block depends on C_t and hence on the business formation block.

$\zeta > 0$: With $\zeta > 0$, $\rho(N_t) > 1$ for $N_t > 1$ and $\rho'(N_t) > 0$, so $w_t^{int} = \rho(N_t)z_t/\mu(N_t)$ depends on N_t. Since N_t is determined in the business formation block, the job creation condition (28) depends on N_{t+1} through w_{t+1}^{int}, breaking separability.

$p_0 > 0$: With $p_0 > 0$, $\Lambda_t = F_\chi(\chi_t^c) < 1$ for $\chi_t^c < \chi_m$, and χ_t^c satisfies the cutoff condition (27) which depends on Y_t^c, N_t, and $\mu(N_t)$ — all determined in the business formation block. Hence $\delta_{e,t} = 1 - (1 - \delta_{t-1})\Lambda_t$ depends on business formation variables. Since $\delta_{e,t}$ enters both laws of motion (6)–(5), the labor market block depends on the business formation block. □

Proof of Proposition 2

Consider the DS-CES equilibrium with $p_0 = 0$ and parameters (ε, f_e) satisfying $f_e = \bar{f}_e/\varepsilon$ for fixed $\bar{f}_e > 0$. Since $p_0 = 0$, $\Lambda_t = 1$ and $\delta_{e,t} = \delta_{t-1}$ throughout. We establish claims (iii), (iv), (i), and (ii) in that order, since (iii) is used in (iv) and (i) validates an assumption used in (iii).

Claim (iii): $w_t^{int} \to z_t$.. Under DS-CES, $\mu = \varepsilon/(\varepsilon-1) \to 1$ and $\rho(N_t) = N_t^{1/(\varepsilon-1)}$. For any N_t satisfying $N_t = O(e^{o(\varepsilon)})$ as $\varepsilon \to \infty$ — a condition weaker than boundedness, verified below — $\ln \rho(N_t) = \ln N_t/(\varepsilon-1) \to 0$, so $\rho(N_t) \to 1$. Hence $w_t^{int} = \rho(N_t)z_t/\mu(N_t) \to z_t$ along any such equilibrium path. □

Claim (iv): N_t drops out of aggregate allocations.. We show that $\{\theta_t, u_t, v_t, C_t, Y_t^c\}$ are determined independently of N_t in the limit.

Labor market block. From claim (iii), $w_t^{int} \to z_t$. Since $p_0 = 0$ gives $\Lambda_t = 1$, the job creation condition (28), flow vacancy value (29), free entry (30), and laws of motion (6)–(5) converge to (31)–(32), which contain no reference to N_t.

Resource constraint. Since $p_0 = 0$, $X_t^c = 0$. Labor devoted to entry satisfies $l_t^e = f_e N_t^e/z_t = O(1/\varepsilon) \to 0$, established in claim (ii). Hence $L_t^c = L_t - l_t^e \to L_t = 1 - u_t$. Output per firm $y_t = z_t L_t^c/N_t \to z_t(1-u_t)/N_t$, and $Y_t^c = \rho(N_t)N_t y_t \to 1 \cdot N_t \cdot z_t(1-u_t)/N_t = z_t(1-u_t)$. The resource constraint becomes $C_t = z_t(1-u_t) - X_t$, independent of N_t.

Since all equations determining $\{\theta_t, u_t, v_t, C_t, Y_t^c\}$ are independent of N_t in the limit, the normalization $\bar{N} = 1$ is a units convention for the measure of product lines — analogous to normalizing household measure to unity — and involves no loss of generality for aggregate allocations. No restriction on $\bar{f}_e$ is required to implement it. □

Claim (i): steady-state value of N_t.. Since aggregate allocations are independent of N_t by claim (iv), the dynamic path of N_t does not affect the proposition's main result. We nonetheless characterize the steady state and local dynamics of N_t to confirm it remains well-defined in the limiting economy.

Substituting $f_e = \bar{f}_e/\varepsilon$ and $\rho(N_t) = N_t^{1/(\varepsilon-1)}$ into the firm LOM (24) with $p_0 = 0$ and taking $\varepsilon \to \infty$, using the approximation $N^{-1/(\varepsilon-1)} = e^{-\ln N/(\varepsilon-1)} \approx 1 - \ln N/(\varepsilon-1)$:

$$N_t \to (1-\delta_{t-1})\left(N_{t-1} + \frac{z_{t-1}(1-u_{t-1})\ln N_{t-1}}{\bar{f}_e}\right) \tag{C.1}$$

This is a nonlinear predetermined variable equation driven by the exogenous processes $\{z_t, \delta_t, u_t\}$, with no forward-looking terms. Its unique positive steady state $\bar{N}$ satisfies:

$$\bar{\delta}\bar{N} = \frac{\bar{z}\bar{L}\ln\bar{N}}{\bar{f}_e} \tag{C.2}$$

where $\bar{L} = 1 - \bar{u}$. The right-hand side is zero at $\bar{N} = 1$, strictly positive and increasing for $\bar{N} > 1$, while the left-hand side is linear with positive slope $\bar{\delta}$. A unique solution $\bar{N} > 1$ exists for any $\bar{\delta}, \bar{f}_e, \bar{z}, \bar{L} > 0$ by the intermediate value theorem and strict monotonicity.

Linearizing (C.1) around $\bar{N}$ gives autoregressive coefficient:

$$\rho_N \equiv (1-\bar{\delta})\left(1+\frac{\bar{z}\bar{L}}{\bar{f}_e\bar{N}}\right) = (1-\bar{\delta})\left(1+\frac{\bar{\delta}}{\ln \bar{N}}\right)$$

where the second equality uses (C.2). Local stability requires $\rho_N < 1$, which holds if and only if $\bar{N} > e^{1-\bar{\delta}}$. For $\bar{\delta} \approx 0.05$, this requires $\bar{N} > e^{0.95} \approx 2.59$ — satisfied for sufficiently small $\bar{f}_e$ given other parameters. When the stability condition holds, N_t follows a well-defined stationary process driven by $\{z_t, \delta_t, u_t\}$, confirming the path satisfies $N_t = O(e^{o(\varepsilon)})$ as required by claim (iii). □

Claim (ii): rates of convergence.. *Firm value*: $\nu_t^f = \rho(N_t)f_e/\mu(N_t) \approx 1 \cdot (\bar{f}_e/\varepsilon) \cdot 1 = O(1/\varepsilon)$.

Investment: At steady state, $N_t^e = \bar{\delta}\bar{N}/(1-\bar{\delta}) = O(1)$, so $\nu_t^f N_t^e = O(1/\varepsilon) \cdot O(1) = O(1/\varepsilon)$.

Labor to entry: $l_t^e = f_e N_t^e/z_t = (\bar{f}_e/\varepsilon) \cdot O(1)/z_t = O(1/\varepsilon)$. □

Proof of Proposition 3

Under clone replacement, $N_t = 1$ and $\mu = 1$ are maintained by construction at all times. We verify that the business formation block drops out and the remaining system reduces to (31)–(32).

Business formation block. Substituting $N_t = 1$ and $\mu = 1$ into the baseline equilibrium definition:

The firm LOM (24): with $N_t = 1$ maintained by construction, this equation is satisfied trivially by the clone replacement assumption and imposes no restriction on other variables.

The business formation Euler (25): with $\rho(1) = 1$, $\mu(1) = 1$, $f_e\rho(N_t) = f_e$, and $(\mu(N_{t+1}) - 1) = 0$, the right-hand side reduces to $(1-\delta_t)\mathbb{E}_t m_{t+1}\Lambda_{t+1}\{f_e\} = f_e \cdot (1-\delta_t)\mathbb{E}_t m_{t+1}$. This is a condition on f_e alone and is satisfied by appropriate choice of f_e given β and $\bar{\delta}$; it imposes no restriction on labor market variables.

The cutoff condition (27): with $\mu = 1$, $(\mu-1)/\mu = 0$, so $\chi_t^c = f_e\rho(N_t)/\mu(N_t) = f_e$. Since clone replacement maintains $N_t = 1$ regardless of profitability, the exit decision is moot; $\chi_t^c = f_e$ is a constant that imposes no restriction on aggregate allocations.

The resource constraint (26): since $p_0 = 0$ under clone replacement (endogenous exit is absent by construction), $X_t^c = 0$. With $\rho(1) = 1$ and $\mu(1) = 1$, $Y_t^c = z_t(1-u_t)$ and $l_t^e = 0$, giving $C_t = z_t(1-u_t) - X_t$, which is (**??**).

Labor market block. Substituting $\rho(N_t) = 1$, $\mu(N_t) = 1$, $\Lambda_t = 1$, and $\delta_{e,t} = \delta_{t-1}$ into (28)–(5) yields (31)–(32) directly. The stochastic discount factor $m_t = \beta(C_t/C_{t-1})^{-\sigma}$ is unchanged. The stochastic processes (7)–(8) are unchanged.

The resulting system (31)–(32) together with $e_t = G(Q_t)$, $m_t = \beta(C_t/C_{t-1})^{-\sigma}$, $C_t = z_t(1-u_t) - X_t$, and stochastic processes (7)–(8) is well-posed: it inherits a unique bounded equilibrium from standard DSGE existence conditions, since it is a special case of the baseline model with $N_t = 1$ fixed, which satisfies the Blanchard-Kahn conditions on the labor market block established in the baseline analysis. $\square$

Appendix C.1. Details of comparative statics

Comment out this section temporarily

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